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Microsoft

DP-700

Implementing Data
Engineering Solutions Using
Microsoft Fabric

Exam Topic: Implement and manage an analytics solution (Testlet 1)**Case Study:**

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When you are ready to answer a question, click the Question button to return to the question.

Overview:

Litware, Inc. is a publishing company that has an online bookstore and several retail bookstores worldwide. Litware also manages an online advertising business for the authors it represents.

Existing Environment. Fabric Environment

Litware has a Fabric workspace named Workspace1. High concurrency is enabled for Workspace1. The company has a data engineering team that uses Python for data processing.

Existing Environment. Data Processing

The retail bookstores send sales data at the end of each business day, while the online bookstore constantly provides logs and sales data to a central enterprise resource planning (ERP) system. Litware implements a medallion architecture by using the following three layers: bronze, silver, and gold. The sales data is ingested from the ERP system as Parquet files that land in the Files folder in a lakehouse. Notebooks are used to transform the files in a Delta table for the bronze and silver layers. The gold layer is in a warehouse that has V-Order disabled. Litware has image files of book covers in Azure Blob Storage. The files are loaded into the Files folder.

Existing Environment. Sales Data

Month-end sales data is processed on the first calendar day of each month. Data that is older than one month never changes.

In the source system, the sales data refreshes every six hours starting at midnight each day.

The sales data is captured in a Dataflow Gen2 dataflow.

When the dataflow runs, new and historical data is captured. The dataflow captures the following fields of the source:

Sales Date
Author
Price
Units
SKU

A table named AuthorSales stores the sales data that relates to each author. The table contains a column

named AuthorEmail. Authors authenticate to a guest Fabric tenant by using their email address.

Existing Environment. Security Groups

Litware has the following security groups:

Sales

Fabric Admins

Streaming Admins

Existing Environment. Performance Issues

Business users perform ad-hoc queries against the warehouse. The business users indicate that reports against the warehouse sometimes run for two hours and fail to load as expected. Upon further investigation, the data engineering team receives the following error message when the reports fail to load: "The SQL query failed while running."

The data engineering team wants to debug the issue and find queries that cause more than one failure. When the authors have new book releases, there is often an increase in sales activity. This increase slows the data ingestion process.

The company's sales team reports that during the last month, the sales data has NOT been up-to-date when they arrive at work in the morning.

Requirements. Planned Changes

Litware recently signed a contract to receive book reviews. The provider of the reviews exposes the data in Amazon Simple Storage Service (Amazon S3) buckets.

Litware plans to manage Search Engine Optimization (SEO) for the authors. The SEO data will be streamed from a REST API.

Requirements. Version Control

Litware plans to implement a version control solution in Fabric that will use GitHub integration and follow the principle of least privilege.

Requirements. Governance Requirements

To control data platform costs, the data platform must use only Fabric services and items. Additional Azure resources must NOT be provisioned.

Requirements. Data Requirements

Litware identifies the following data requirements:

Process the SEO data in near-real-time (NRT).

Make the book reviews available in the lakehouse without making a copy of the data.

When a new book cover image arrives in the Files folder, process the image as soon as possible.

QUESTION: 1

You need to ensure that processes for the bronze and silver layers run in isolation.

How should you configure the Apache Spark settings?

- A. Disable high concurrency.
- B. Create a custom pool.
- C. Modify the number of executors.
- D. Set the default environment.

Answer(s): B

Explanation:

Isolated Compute

The Isolated Compute option provides more security to Spark compute resources from untrusted services by dedicating the physical compute resource to a single customer. Isolated compute option is best suited for workloads that require a high degree of isolation from other customer's workloads for reasons that

include meeting compliance and regulatory requirements. The Isolate Compute option is only available with the XXXLarge (80 vCPU / 504 GB) node size and only available in the following regions. *The isolated compute option can be enabled or disabled after pool creation although the instance might need to be restarted*.

Scenario:

Existing Environment. Data Processing

Litware implements a medallion architecture by using the following three layers: bronze, silver, and gold. The sales data is ingested from the ERP system as Parquet files that land in the Files folder in a lakehouse.

Notebooks are used to transform the files in a Delta table for the bronze and silver layers. The gold layer is in a warehouse that has V-Order disabled.

QUESTION: 2

DRAG DROP (Drag and Drop is not supported)

You need to ensure that the authors can see only their respective sales data.

How should you complete the statement? To answer, drag the appropriate values the correct targets.

Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

Note: Each correct selection is worth one point.

Select and Place:

Values

AuthorSales

AuthorEmail

AuthorSales.AuthorEmail

BLOCK

FILTER

INLINE

SCHEMABINDING

USER_NAME()

Answer Area

```
CREATE FUNCTION dbo.tvf_rlspredicate(@Author AS varchar(50))
```

```
RETURNS TABLE
```

```
WITH
```

```
AS
```

```
RETURN SELECT 1 AS tvf_rlspredicate_result
```

```
WHERE @Author =
```

```
GO
```

```
CREATE SECURITY POLICY RLSFilter
```

```
ADD FILTER PREDICATE Security.tvf_rlspredicate(AuthorEmail)
```

```
ON
```

```
WITH (STATE = ON)
```

A. See Explanation section for answer.

Answer(s): A

Explanation:

Values

AuthorSales

AuthorEmail

AuthorSales.AuthorEmail

BLOCK

FILTER

INLINE

SCHEMABINDING

USER_NAME()

Answer Area

```

CREATE FUNCTION dbo.tvf_rlspredicate(@Author AS varchar(50))
    RETURNS TABLE
WITH SCHEMABINDING
AS
    RETURN SELECT 1 AS tvf_rlspredicate_result
WHERE @Author = USER_NAME()
GO

CREATE SECURITY POLICY RLSFilter
ADD FILTER PREDICATE Security.tvf_rlspredicate(AuthorEmail)
ON AuthorSales
WITH (STATE = ON)

```

Box 1: SCHEMABINDING

CREATE FUNCTION (Azure Synapse Analytics and Microsoft Fabric) Creates a user-defined function (UDF) in Azure Synapse Analytics, Analytics Platform System (PDW), or Microsoft Fabric. A user-defined function is a Transact-SQL routine that accepts parameters, performs an action, such as a complex calculation, and returns the result of that action as a value.

In Microsoft Fabric and serverless SQL pools in Azure Synapse Analytics, CREATE FUNCTION can create inline table-valued functions but not scalar functions. User-defined table-valued functions (TVFs) return a table data type.

Inline table-valued function syntax

-- Transact-SQL Inline Table-Valued Function Syntax

-- Preview in dedicated SQL pools in Azure Synapse Analytics -- Available in the serverless SQL pools in Azure Synapse Analytics and Microsoft Fabric CREATE FUNCTION [schema_name.] function_name

([{ @parameter_name [AS] parameter_data_type

[= default] }

[,...n]

]

)

RETURNS TABLE

[WITH SCHEMABINDING]

[AS]

RETURN [() select_stmt []]

[;]

--

SCHEMABINDING

Specifies that the function is bound to the database objects that it references.

When SCHEMABINDING is specified, the base objects cannot be modified in a way that would affect the function definition. The function definition itself must first be modified or dropped to remove dependencies on the object that is to be modified.

Box 2: USER_NAME()

USER_NAME (Transact-SQL)

Returns a database user name from a specified identification number, or the current user name.

Box 3: AuthorSales

We specify the table name, table_schema_name.table_name below.

Scenario:

Litware also manages an online advertising business for the authors it represents.

Existing Environment. Sales Data

A table named AuthorSales stores the sales data that relates to each author. The table contains a column named AuthorEmail. Authors authenticate to a guest Fabric tenant by using their email address.

Note: CREATE SECURITY POLICY (Transact-SQL)

Create a security policy

The following syntax creates a security policy with a filter predicate for the dbo.Customer table, and leaves the security policy disabled.

SQL

```
CREATE SECURITY POLICY [FederatedSecurityPolicy]
```

```
ADD FILTER PREDICATE [rls].[fn_securitypredicate]([CustomerId]) ON [dbo].[Customer];
```

```
--
```

```
table_schema_name.table_name
```

Is the target table to which the security predicate will be applied. Multiple disabled security policies can target a single table for a particular DML operation, but only one can be enabled at any given time.

Full Syntax:

```
CREATE SECURITY POLICY [schema_name. ] security_policy_name
```

```
{ ADD [ FILTER | BLOCK ] } PREDICATE tvf_schema_name.security_predicate_function_name ( {  
column_name | expression } [ , ...n ] ) ON table_schema_name. table_name [ <block_dml_operation> ] , [ ,  
...n ]
```

```
[ WITH ( STATE = { ON | OFF } [ , ] [ SCHEMABINDING = { ON | OFF } ] ) ] [ NOT FOR REPLICATION ]  
[ ; ]
```

Exam Topic: Implement and manage an analytics solution (Testlet 2)**Case Study:**

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Overview:

- Company Overview

Contoso, Ltd. is an online retail company that wants to modernize its analytics platform by moving to Fabric. The company plans to begin using Fabric for marketing analytics.

Overview:

- IT Structure

The company's IT department has a team of data analysts and a team of data engineers that use analytics systems.

The data engineers perform the ingestion, transformation, and loading of data. They prefer to use Python or SQL to transform the data.

The data analysts query data and create semantic models and reports. They are qualified to write queries in Power Query and T-SQL.

Existing Environment. Fabric

Contoso has an F64 capacity named Cap1. All Fabric users are allowed to create items.

Contoso has two workspaces named WorkspaceA and WorkspaceB that currently use Pro license mode.

Existing Environment. Source Systems

Contoso has a point of sale (POS) system named POS1 that uses an instance of SQL Server on Azure Virtual Machines in the same Microsoft Entra tenant as Fabric. The host virtual machine is on a private virtual network that has public access blocked. POS1 contains all the sales transactions that were processed on the company's website.

The company has a software as a service (SaaS) online marketing app named MAR1. MAR1 has seven entities. The entities contain data that relates to email open rates and interaction rates, as well as website interactions. The data can be exported from MAR1 by calling REST APIs. Each entity has a different endpoint.

Contoso has been using MAR1 for one year. Data from prior years is stored in Parquet files in an Amazon Simple Storage Service (Amazon S3) bucket. There are 12 files that range in size from 300 MB to 900 MB and relate to email interactions.

Existing Environment. Product Data

POS1 contains a product list and related data. The data comes from the following three tables:

Products

ProductCategories

ProductSubcategories

In the data, products are related to product subcategories, and subcategories are related to product categories.

Existing Environment. Azure

Contoso has a Microsoft Entra tenant that has the following mail-enabled security groups:

DataAnalysts: Contains the data analysts

DataEngineers: Contains the data engineers

Contoso has an Azure subscription.

The company has an existing Azure DevOps organization and creates a new project for repositories that relate to Fabric.

Existing Environment. User Problems

The VP of marketing at Contoso requires analysis on the effectiveness of different types of email content. It typically takes a week to manually compile and analyze the data. Contoso wants to reduce the time to less than one day by using Fabric.

The data engineering team has successfully exported data from MAR1. The team experiences transient connectivity errors, which causes the data exports to fail.

Requirements. Planned Changes

Contoso plans to create the following two lakehouses:

Lakehouse1: Will store both raw and cleansed data from the sources

Lakehouse2: Will serve data in a dimensional model to users for analytical queries

Additional items will be added to facilitate data ingestion and transformation.

Contoso plans to use Azure Repos for source control in Fabric.

Requirements. Technical Requirements

The new lakehouses must follow a medallion architecture by using the following three layers: bronze, silver, and gold. There will be extensive data cleansing required to populate the MAR1 data in the silver layer, including deduplication, the handling of missing values, and the standardizing of capitalization. Each layer must be fully populated before moving on to the next layer. If any step in populating the lakehouses fails, an email must be sent to the data engineers.

Data imports must run simultaneously, when possible.

The use of email data from the Amazon S3 bucket must meet the following requirements:

Minimize egress costs associated with cross-cloud data access.

Prevent saving a copy of the raw data in the lakehouses.

Items that relate to data ingestion must meet the following requirements:

The items must be source controlled alongside other workspace items.

Ingested data must land in the bronze layer of Lakehouse1 in the Delta format.

No changes other than changes to the file formats must be implemented before the data lands in the bronze layer.

Development effort must be minimized and a built-in connection must be used to import the source data.

In the event of a connectivity error, the ingestion processes must attempt the connection again.

Lakehouses, data pipelines, and notebooks must be stored in WorkspaceA. Semantic models, reports, and dataflows must be stored in WorkspaceB.

Once a week, old files that are no longer referenced by a Delta table log must be removed.

Requirements. Data Transformation

In the POS1 product data, ProductID values are unique. The product dimension in the gold layer must include only active products from product list. Active products are identified by an IsActive value of 1.

Some product categories and subcategories are NOT assigned to any product. They are NOT analytically relevant and must be omitted from the product dimension in the gold layer.

Requirements. Data Security

Security in Fabric must meet the following requirements:

The data engineers must have read and write access to all the lakehouses, including the underlying files.

The data analysts must only have read access to the Delta tables in the gold layer.

The data analysts must NOT have access to the data in the bronze and silver layers.

The data engineers must be able to commit changes to source control in WorkspaceA.

QUESTION: 3

You need to ensure that the data analysts can access the gold layer lakehouse.

What should you do?

A. Add the DataAnalyst group to the Viewer role for WorkspaceA.

B. Share the lakehouse with the DataAnalysts group and grant the Build reports on the default semantic model permission.

C. Share the lakehouse with the DataAnalysts group and grant the Read all SQL Endpoint data permission.

D. Share the lakehouse with the DataAnalysts group and grant the Read all Apache Spark permission.

Answer(s): C**Explanation:**

Data Analysts' Access Requirements must only have read access to the Delta tables in the gold layer and not have access to the bronze and silver layers.

The gold layer data is typically queried via SQL Endpoints. Granting the Read all SQL Endpoint data permission allows data analysts to query the data using familiar SQL-based tools while restricting access to the underlying files.

QUESTION: 4

You need to ensure that WorkspaceA can be configured for source control.

Which two actions should you perform? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. From Tenant setting, set Users can synchronize workspace items with their Git repositories to Enabled.
- B. From Tenant setting, set Users can sync workspace items with GitHub repositories to Enabled.
- C. Configure WorkspaceA to use a Premium Per User (PPU) license.
- D. Assign WorkspaceA to Cap1.

Answer(s): A, D

Explanation:

A F64 capacity is a Microsoft Fabric Capacity (Pay-as-you-go or Reservation).

Note: Microsoft Fabric, Get started with Git integration

Prerequisites

To integrate Git with your Microsoft Fabric workspace, you need to set up the following prerequisites for both Fabric and Git.

Fabric prerequisites

To access the Git integration feature, you need a Fabric capacity. A Fabric capacity is required to use all supported Fabric items [D]. If you don't have one yet, sign up for a free trial. Customers that already have a Power BI Premium capacity, can use that capacity, but keep in mind that certain Power BI SKUs only support Power BI items.

In addition, the following tenant switches must be enabled from the Admin portal:

* Users can create Fabric items

*-> Users can synchronize workspace items with their Git repositories [A, not B] For GitHub users only:

Users can synchronize workspace items with GitHub repositories Scenario:

Existing Environment. Fabric

Contoso has an F64 capacity named Cap1 [D]. All Fabric users are allowed to create items.

Contoso has two workspaces named WorkspaceA and WorkspaceB that currently use Pro license mode.

Requirements. Planned Changes

Contoso plans to use Azure Repos for source control in Fabric.

Requirements. Technical Requirements

Items that relate to data ingestion must meet the following requirements:

*-> The items must be source controlled alongside other workspace items.

Requirements. Data Security

Security in Fabric must meet the following requirements:

*-> The data engineers must be able to commit changes to source control in WorkspaceA.

Exam Topic: Implement and manage an analytics solution (Question Set 3)**QUESTION: 5**

You have a Fabric workspace.

You have semi-structured data.

You need to read the data by using T-SQL, KQL, and Apache Spark. The data will only be written by using Spark.

What should you use to store the data?

- A. a lakehouse
- B. an eventhouse

- C. a datamart
- D. a warehouse

Answer(s): A

Explanation:

A lakehouse is the best option for storing semi-structured data when you need to read it using T-SQL, KQL, and Apache Spark. A lakehouse combines the flexibility of a data lake (which can handle semi-structured and unstructured data) with the performance features of a data warehouse. It allows data to be written using Apache Spark and can be queried using different technologies such as T-SQL (for SQL-based querying), KQL (Kusto Query Language for querying), and Apache Spark (for distributed processing). This solution is ideal when dealing with semi-structured data and requiring a versatile querying approach.

QUESTION: 6

You have a Fabric workspace that contains a warehouse named Warehouse1.

You have an on-premises Microsoft SQL Server database named Database1 that is accessed by using an on- premises data gateway.

You need to copy data from Database1 to Warehouse1.

Which item should you use?

- A. a Dataflow Gen1 dataflow
- B. a data pipeline
- C. a KQL queryset
- D. a notebook

Answer(s): B

Explanation:

To copy data from an on-premises Microsoft SQL Server database (Database1) to a warehouse (Warehouse1) in Microsoft Fabric, the best option is to use a data pipeline. A data pipeline in Fabric allows for the orchestration of data movement, from source to destination, using connectors, transformations, and scheduled workflows. Since the data is being transferred from an on-premises database and requires the use of a data gateway, a data pipeline provides the appropriate framework to facilitate this data movement efficiently and reliably.

QUESTION: 7

You have a Fabric workspace that contains a warehouse named Warehouse1.

You have an on-premises Microsoft SQL Server database named Database1 that is accessed by using an on- premises data gateway.

You need to copy data from Database1 to Warehouse1.

Which item should you use?

- A. an Apache Spark job definition
- B. a data pipeline
- C. a Dataflow Gen1 dataflow
- D. an eventstream

Answer(s): B

Explanation:

To copy data from an on-premises Microsoft SQL Server database (Database1) to a warehouse (Warehouse1) in Fabric, a data pipeline is the most appropriate tool. A data pipeline in Fabric is designed to move data between various data sources and destinations, including on-premises databases like SQL Server, and cloud- based storage like Fabric warehouses. The data pipeline can handle the connection through an on-premises data gateway, which is required to access on-premises data. This solution facilitates the orchestration of data movement and transformations if needed.

QUESTION: 8

You have a Fabric F32 capacity that contains a workspace. The workspace contains a warehouse named DW1 that is modelled by using MD5 hash surrogate keys.

DW1 contains a single fact table that has grown from 200 million rows to 500 million rows during the past year.

You have Microsoft Power BI reports that are based on Direct Lake. The reports show year-over-year values.

Users report that the performance of some of the reports has degraded over time and some visuals show errors.

You need to resolve the performance issues. The solution must meet the following requirements:

Provide the best query performance.

Minimize operational costs.

Which should you do?

- A. Change the MD5 hash to SHA256.
- B. Increase the capacity.
- C. Enable V-Order.
- D. Modify the surrogate keys to use a different data type.
- E. Create views.

Answer(s): B

Explanation:

Correct:

* Increase the capacity.

We need to upgrade the capacity.

Direct Lake in Fabric F32 Capacity supports tables up to 300 million rows. F64 can handle 500 million rows.

Incorrect:

* Change the MD5 hash to SHA256.

* Create views.

* Disable V-Order on the warehouse.

* Enable V-Order.

* Modify the surrogate keys to use a different data type.

QUESTION: 9

HOTSPOT (Drag and Drop is not supported)

You have a Fabric workspace that contains a warehouse named DW1. DW1 contains the following tables and columns.

Table name	Column name	Description
SalesOrderDetail	ProductID	Contains the product ID of the ordered product
SalesOrderDetail	ModifiedDate	Contains the date of an order
SalesOrderDetail	OrderQty	Contains the order quantity
Product	ProductID	Contains the unique ID of a product
Product	Name	Contains a product name

You need to create an output that presents the summarized values of all the order quantities by year and product. The results must include a summary of the order quantities at the year level for all the products.

How should you complete the code? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

	▼	(SO.ModifiedDate) AS OrderDate
SELECT CAST		
SELECT CONVERT		
SELECT YEAR		
,P.Name AS ProductName		
,SUM(SO.OrderQty) AS OrderQty		
FROM [dbo].[SalesOrderDetail] SO		
INNER JOIN [dbo].[Product] P		
ON P.ProductID = SO.ProductID		
GROUP BY		
▼		
CUBE(YEAR(SO.ModifiedDate), P.Name)		
GROUPING SETS ((YEAR(SO.ModifiedDate), P.Name), (YEAR(SO.ModifiedDate)))		
ROLLUP(YEAR(SO.ModifiedDate), P.Name)		
YEAR(SO.ModifiedDate), P.Name		
ORDER BY OrderDate		

A. See Explanation section for answer.

Answer(s): A**Explanation:**

Answer Area

	(SO.ModifiedDate) AS OrderDate
SELECT CAST	
SELECT CONVERT	
SELECT YEAR	

```

,P.Name AS ProductName
,SUM(SO.OrderQty) AS OrderQty
FROM [dbo].[SalesOrderDetail] SO
INNER JOIN [dbo].[Product] P
ON P.ProductID = SO.ProductID
GROUP BY

```

CUBE(YEAR(SO.ModifiedDate), P.Name)	
GROUPING SETS ((YEAR(SO.ModifiedDate), P.Name), (YEAR(SO.ModifiedDate)))	
ROLLUP(YEAR(SO.ModifiedDate), P.Name)	
YEAR(SO.ModifiedDate), P.Name	

```

ORDER BY OrderDate

```

Summarize by year and product: The query needs to group data by both year and product name. Include a summary of order quantities at the year level: This is achieved using grouping mechanisms like ROLLUP or CUBE.

SELECT YEAR(SO.ModifiedDate)

Since we need to summarize the data by year, extracting the year from the ModifiedDate column using YEAR (SO.ModifiedDate) is the correct choice. Other options like CAST or CONVERT do not specifically extract the year.

ROLLUP(YEAR(SO.ModifiedDate), P.Name)

The ROLLUP function creates subtotals for each grouping combination. In this case:

It will group by YEAR(SO.ModifiedDate) and P.Name (product name).

It will also include a summary for all products for each year, which meets the requirement of summarizing order quantities at the year level for all products.

```

SELECT
    YEAR(SO.ModifiedDate) AS OrderDate,
    P.Name AS ProductName,
    SUM(SO.OrderQty) AS OrderQty
FROM
    [dbo].[SalesOrderDetail] SO
INNER JOIN
    [dbo].[Product] P
ON
    P.ProductID = SO.ProductID
GROUP BY
    ROLLUP(YEAR(SO.ModifiedDate), P.Name)
ORDER BY
    OrderDate;

```

QUESTION: 10

You have a Fabric workspace that contains a lakehouse named Lakehouse1. Data is ingested into Lakehouse1 as one flat table. The table contains the following columns.

Name	Description
TransactionID	Contains a unique ID for each transaction
Date	Contains the date of a transaction
ProductID	Contains a unique ID for each product
ProductColor	Contains a descriptive attribute that describes the color of each product
ProductName	Contains a unique name for each product
SalesAmount	Contains the sales amount of a transaction

You plan to load the data into a dimensional model and implement a star schema. From the original flat table, you create two tables named FactSales and DimProduct. You will track changes in DimProduct. You need to prepare the data.

Which three columns should you include in the DimProduct table? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. Date
- B. ProductName
- C. ProductColor
- D. TransactionID
- E. SalesAmount
- F. ProductID

Answer(s): B, C, F

Explanation:

In a star schema, the DimProduct table serves as a dimension table that contains descriptive attributes about products. It will provide context for the FactSales table, which contains transactional data. The following columns should be included in the DimProduct table:

1. ProductName: The ProductName is an important descriptive attribute of the product, which is needed for analysis and reporting in a dimensional model.
2. ProductColor: ProductColor is another descriptive attribute of the product. In a star schema, it makes sense to include attributes like color in the dimension table to help categorize products in the analysis.
3. ProductID: ProductID is the primary key for the DimProduct table, which will be used to join the FactSales table to the product dimension. It's essential for uniquely identifying each product in the model.

QUESTION: 11

You have a Fabric workspace named Workspace1 that contains a notebook named Notebook1.

In Workspace1, you create a new notebook named Notebook2.

You need to ensure that you can attach Notebook2 to the same Apache Spark session as Notebook1. What should you do?

- A. Enable high concurrency for notebooks.
- B. Enable dynamic allocation for the Spark pool.
- C. Change the runtime version.
- D. Increase the number of executors.

Answer(s): A

Explanation:

To ensure that Notebook2 can attach to the same Apache Spark session as Notebook1, you need to enable high concurrency for notebooks. High concurrency allows multiple notebooks to share a Spark session, enabling them to run within the same Spark context and thus share resources like cached data, session state, and compute capabilities. This is particularly useful when you need notebooks to run in sequence or together while leveraging shared resources.

QUESTION: 12

You have a Fabric workspace named Workspace1 that contains a lakehouse named Lakehouse1.

Lakehouse1 contains the following tables:

Orders

Customer

Employee

The Employee table contains Personally Identifiable Information (PII).

A data engineer is building a workflow that requires writing data to the Customer table, however, the user does NOT have the elevated permissions required to view the contents of the Employee table.

You need to ensure that the data engineer can write data to the Customer table without reading data from the Employee table.

Which three actions should you perform? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. Share Lakehouse1 with the data engineer.
- B. Assign the data engineer the Contributor role for Workspace2.
- C. Assign the data engineer the Viewer role for Workspace2.
- D. Assign the data engineer the Contributor role for Workspace1.
- E. Migrate the Employee table from Lakehouse1 to Lakehouse2.
- F. Create a new workspace named Workspace2 that contains a new lakehouse named Lakehouse2.
- G. Assign the data engineer the Viewer role for Workspace1.

Answer(s): D, E, F

QUESTION: 13

You have a Fabric warehouse named DW1. DW1 contains a table that stores sales data and is used by multiple sales representatives.

You plan to implement row-level security (RLS).

You need to ensure that the sales representatives can see only their respective data.

Which warehouse object do you require to implement RLS?

- A. STORED PROCEDURE
- B. CONSTRAINT
- C. SCHEMA
- D. FUNCTION

Answer(s): D

Explanation:

To implement Row-Level Security (RLS) in a Fabric warehouse, you need to use a function that defines the security logic for filtering the rows of data based on the user's identity or role. This function can be used in conjunction with a security policy to control access to specific rows in a table.

In the case of sales representatives, the function would define the filtering criteria (e.g., based on a column such as SalesRepID or SalesRepName), ensuring that each representative can only see their respective data.

QUESTION: 14

HOTSPOT (Drag and Drop is not supported)

You have a Fabric workspace named Workspace1_DEV that contains the following items:

10 reports

Four notebooks

Three lakehouses

Two data pipelines

Two Dataflow Gen1 dataflows

Three Dataflow Gen2 dataflows

Five semantic models that each has a scheduled refresh policy

You create a deployment pipeline named Pipeline1 to move items from Workspace1_DEV to a new workspace named Workspace1_TEST.

You deploy all the items from Workspace1_DEV to Workspace1_TEST.

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Data from the semantic models will be deployed to the target stage.	☐	☐
The Dataflow Gen1 dataflows will be deployed to the target stage.	☐	☐
The scheduled refresh policies will be deployed to the target stage.	☐	☐

A. See Explanation section for answer.

Answer(s): A
Explanation:
 Answer Area

Statements	Yes	No
Data from the semantic models will be deployed to the target stage.	☐	☒
The Dataflow Gen1 dataflows will be deployed to the target stage.	☒	☐
The scheduled refresh policies will be deployed to the target stage.	☐	☒

Explanation:

Data from the semantic models will be deployed to the target stage - No In a deployment pipeline, data from semantic models (such as cached data) is not typically deployed along with the models themselves. While the semantic models themselves (including structure and definitions) will be deployed, the actual data (e.g., the results of the model refreshes) will not be automatically transferred. You would need to handle data refreshes separately after the semantic models are deployed.

The Dataflow Gen1 dataflows will be deployed to the target stage - Yes Dataflows, including Dataflow Gen1 dataflows, are part of the deployment pipeline and will be deployed to the target stage. These dataflows are part of the solution being deployed, and the pipeline ensures their migration to the target workspace (Workspace1_TEST).

The scheduled refresh policies will be deployed to the target stage - No While the scheduled refresh policies for semantic models are part of the configuration, they are not automatically deployed through the pipeline. Deployment pipelines typically move content, such as reports, notebooks, and dataflows, but scheduled refresh settings are not automatically transferred with the deployment. You would need to manually configure the refresh policies in the new workspace (Workspace1_TEST) after the deployment.

QUESTION: 15

You have a Fabric deployment pipeline that uses three workspaces named Dev, Test, and Prod. You need to deploy an eventhouse as part of the deployment process. What should you use to add the eventhouse to the deployment process?

- A. GitHub Actions
- B. a deployment pipeline
- C. an Azure DevOps pipeline

Answer(s): C

Explanation:

Correct:

* an Azure DevOps pipeline

Incorrect:

* a deployment pipeline

* an eventstream

* GitHub Actions

QUESTION: 16

You have a Fabric workspace named Workspace1 that contains a warehouse named Warehouse1. You plan to deploy Warehouse1 to a new workspace named Workspace2.

As part of the deployment process, you need to verify whether Warehouse1 contains invalid references. The solution must minimize development effort. What should you use?

- A. a database project
- B. a deployment pipeline
- C. a Python script
- D. a T-SQL script

Answer(s): B

Explanation:

A deployment pipeline in Fabric allows you to deploy assets like warehouses, datasets, and reports between different workspaces (such as from Workspace1 to Workspace2). One of the key features of a deployment pipeline is the ability to check for invalid references before deployment. This can help identify issues with assets, such as broken links or dependencies, ensuring the deployment is successful without introducing errors. This is the most efficient way to verify references and manage the deployment with minimal development effort.

QUESTION: 17

You have a Fabric workspace that contains a Real-Time Intelligence solution and an eventhouse. Users report that from OneLake file explorer, they cannot see the data from the eventhouse. You enable OneLake availability for the eventhouse. What will be copied to OneLake?

- A. only data added to new databases that are added to the eventhouse
- B. only the existing data in the eventhouse
- C. no data
- D. both new data and existing data in the eventhouse
- E. only new data added to the eventhouse

Answer(s): E

QUESTION: 18

You have a Fabric workspace named Workspace1. You plan to integrate Workspace1 with Azure DevOps. You will use a Fabric deployment pipeline named deployPipeline1 to deploy items from Workspace1 to higher environment workspaces as part of a medallion architecture. You will run deployPipeline1 by using an API call from an Azure DevOps pipeline. You need to configure API authentication between Azure DevOps and Fabric. Which type of authentication should you use?

- A. service principal
- B. Microsoft Entra username and password
- C. managed private endpoint
- D. workspace identity

Answer(s): A

Explanation:

When integrating Azure DevOps with Fabric (Workspace1), using a service principal is the recommended authentication method. A service principal provides a way for applications (such as an Azure DevOps pipeline) to authenticate and interact with resources securely. It allows Azure DevOps to authenticate API calls to Fabric without requiring direct user credentials. This method is ideal for automating tasks such as deploying items through a Fabric deployment pipeline.

QUESTION: 19

You have a Google Cloud Storage (GCS) container named storage1 that contains the files shown in the following table.

Name	Size
ProductFile.parquet	8 MB
StoreFile.json	500 MB
TripsFile.csv	99 MB

You have a Fabric workspace named Workspace1 that has the cache for shortcuts enabled. Workspace1 contains a lakehouse named Lakehouse1. Lakehouse1 has the shortcuts shown in the following table.

Name	Source	Last accessed
Products	ProductFile	12 hours ago
Stores	StoreFile	4 hours ago
Trips	TripsFile	48 hours ago

You need to read data from all the shortcuts.

Which shortcuts will retrieve data from the cache?

- A. Stores only
- B. Products only
- C. Stores and Products only
- D. Products, Stores, and Trips
- E. Trips only
- F. Products and Trips only

Answer(s): C

Explanation:

When reading data from shortcuts in Fabric (in this case, from a lakehouse like Lakehouse1), the cache for shortcuts helps by storing the data locally for quick access. The last accessed timestamp and the cache expiration rules determine whether data is fetched from the cache or from the source (Google Cloud Storage, in this case).

Products: The ProductFile.parquet was last accessed 12 hours ago. Since the cache has data available for up to 12 hours, it is likely that this data will be retrieved from the cache, as it hasn't been too long since it was last accessed.

Stores: The StoreFile.json was last accessed 4 hours ago, which is within the cache retention period. Therefore, this data will also be retrieved from the cache.

Trips: The TripsFile.csv was last accessed 48 hours ago. Given that it's outside the typical caching window (assuming the cache has a maximum retention period of around 24 hours), it would not be retrieved from the cache. Instead, it will likely require a fresh read from the source.

QUESTION: 20

You have a Fabric workspace named Workspace1 that contains an Apache Spark job definition named Job1.

You have an Azure SQL database named Source1 that has public internet access disabled.

You need to ensure that Job1 can access the data in Source1.

What should you create?

- A. an on-premises data gateway
- B. a managed private endpoint
- C. an integration runtime
- D. a data management gateway

Answer(s): B

Explanation:

To allow Job1 in Workspace1 to access an Azure SQL database (Source1) with public internet access disabled, you need to create a managed private endpoint. A managed private endpoint is a secure, private connection that enables services like Fabric (or other Azure services) to access resources such as databases, storage accounts, or other services within a virtual network (VNet) without requiring public internet access.

This approach maintains the security and integrity of your data while enabling access to the Azure SQL database.

QUESTION: 21

You have an Azure Data Lake Storage Gen2 account named storage1 and an Amazon S3 bucket named storage2.

You have the Delta Parquet files shown in the following table.

Name	Stored in	Size	Description
ProductFile	storage1	50 MB	Contains a list of products and their details
TripsFile	storage2	2 GB	Contains one month's worth of taxi trip data
StoreFile	storage2	25 MB	Contains a list of stores and their addresses

You have a Fabric workspace named Workspace1 that has the cache for shortcuts enabled. Workspace1 contains a lakehouse named Lakehouse1. Lakehouse1 has the following shortcuts:

A shortcut to ProductFile aliased as Products

A shortcut to StoreFile aliased as Stores

A shortcut to TripsFile aliased as Trips

The data from which shortcuts will be retrieved from the cache?

- A. Trips and Stores only
- B. Products and Store only
- C. Stores only
- D. Products only
- E. Products, Stores, and Trips

Answer(s): B

Explanation:

When the cache for shortcuts is enabled in Fabric, the data retrieval is governed by the caching behavior, which generally retains data for a specific period after it was last accessed. The data from the shortcuts will be retrieved from the cache if the data is stored in locations that support caching. Here's a breakdown based on the data's location:

Products: The ProductFile is stored in Azure Data Lake Storage Gen2 (storage1). Since Azure Data Lake is a supported storage system in Fabric and the file is relatively small (50 MB), this data is most likely cached and can be retrieved from the cache.

Stores: The StoreFile is stored in Amazon S3 (storage2), and even though it is stored in a different cloud provider, Fabric can cache data from Amazon S3 if caching is enabled. This data (25 MB) is likely cached and retrievable.

Trips: The TripsFile is stored in Amazon S3 (storage2) and is significantly larger (2 GB) compared to the other files.

While Fabric can cache data from Amazon S3, the larger size of the file (2 GB) may exceed typical cache sizes or retention windows, causing this file to likely be retrieved directly from the source instead of the cache.

QUESTION: 22

HOTSPOT (Drag and Drop is not supported)

You have a Fabric workspace named Workspace1 that contains the items shown in the following table.

Name	Type
Notebook1	Notebook
Notebook2	Notebook
Lakehouse1	Lakehouse
Pipeline1	Data pipeline
Model1	Semantic model

For Model1, the Keep your Direct Lake data up to date option is disabled.

You need to configure the execution of the items to meet the following requirements:

Notebook1 must execute every weekday at 8:00 AM.

Notebook2 must execute when a file is saved to an Azure Blob Storage container.

Model1 must refresh when Notebook1 has executed successfully.

How should you orchestrate each item? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Notebook1: ▼

- Add Notebook1 to an Apache Spark job definition.
- Add Notebook1 to Pipeline1.
- From Real-Time hub, configure the execution of Notebook1.

Notebook2: ▼

- Add Notebook2 to an Apache Spark job definition.
- Add Notebook2 to Pipeline1.
- From Real-Time hub, configure the execution of Notebook2.

Pipeline1: ▼

- Add Pipeline1 to an Apache Spark job definition.
- Configure the execution of Pipeline1 by using a schedule.
- From Real-Time hub, configure the execution of Pipeline1.

Model1: ▼

- Add Model1 to Pipeline1.
- From Real-Time hub, configure Model1 to refresh.
- Set Keep your Direct Lake data up to date to On.

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

Notebook1:

Notebook2:

Pipeline1:

Model1:

Explanation:

Notebook1 - Add Notebook1 to Pipeline1.

To schedule Notebook1 to run every weekday at 8:00 AM, the best way to orchestrate it is by adding it to Pipeline1 and then configuring a schedule for the pipeline execution. This allows you to leverage the scheduling capabilities of pipelines in Fabric, ensuring that Notebook1 runs at the desired time.

Notebook2 - From Real-Time hub, configure the execution of Notebook2. Since Notebook2 needs to execute when a file is saved to an Azure Blob Storage container, the best way to trigger this is by using Real-Time hub. The Real-Time hub is designed for event-driven scenarios like this, where an external event (e.g., file saved to a blob) can trigger the execution of a notebook.

Pipeline1 - Configure the execution of Pipeline1 by using a schedule. To ensure Model1 refreshes when Notebook1 has executed successfully, you can orchestrate the execution flow by adding Notebook1 to Pipeline1 and configuring the pipeline to execute on a schedule (such as when Notebook1 runs). You can also use dependencies within the pipeline to ensure Model1 only refreshes after Notebook1 executes.

Model1 - Add Model1 to Pipeline1.

To ensure Model1 refreshes when Notebook1 executes, you need to add Model1 to Pipeline1 and configure the execution within the pipeline. By configuring the pipeline so that Model1 refreshes after Notebook1 has successfully executed, you establish the correct dependency. Additionally, ensure that the Keep your Direct Lake data up to date option is disabled since the refresh is controlled by Pipeline1.

QUESTION: 23

Your company has a sales department that uses two Fabric workspaces named Workspace1 and Workspace2.

The company decides to implement a domain strategy to organize the workspaces.
 You need to ensure that a user can perform the following tasks:
 Create a new domain for the sales department.
 Create two subdomains: one for the east region and one for the west region.
 Assign Workspace1 to the east region subdomain.
 Assign Workspace2 to the west region subdomain.
 The solution must follow the principle of least privilege.
 Which role should you assign to the user?

- A. workspace Admin
- B. domain admin
- C. domain contributor
- D. Fabric admin

Answer(s): D

QUESTION: 24

You have a Fabric workspace named Workspace1 that contains a warehouse named DW1 and a data pipeline named Pipeline1.
 You plan to add a user named User3 to Workspace1.
 You need to ensure that User3 can perform the following actions:
 View all the items in Workspace1.
 Update the tables in DW1.
 The solution must follow the principle of least privilege.
 You already assigned the appropriate object-level permissions to DW1.
 Which workspace role should you assign to User3?

- A. Admin
- B. Member
- C. Viewer
- D. Contributor

Answer(s): D

Explanation:

To ensure User3 can view all items in Workspace1 and update the tables in DW1, the most appropriate workspace role to assign is the Contributor role. This role allows User3 to:

1. View all items in Workspace1: The Contributor role provides the ability to view all objects within the workspace, such as data pipelines, warehouses, and other resources.
2. Update the tables in DW1: The Contributor role allows User3 to modify or update resources within the workspace, including the tables in DW1, assuming that appropriate object-level permissions are set for the warehouse.

This role adheres to the principle of least privilege, as it provides the necessary permissions without granting broader administrative rights.

QUESTION: 25

You have a Fabric capacity that contains a workspace named Workspace1. Workspace1 contains a lakehouse named Lakehouse1, a data pipeline, a notebook, and several Microsoft Power BI reports.
 A user named User1 wants to use SQL to analyze the data in Lakehouse1.
 You need to configure access for User1. The solution must meet the following requirements:
 Provide User1 with read access to the table data in Lakehouse1.
 Prevent User1 from using Apache Spark to query the underlying files in Lakehouse1.
 Prevent User1 from accessing other items in Workspace1.
 What should you do?

- A. Share Lakehouse1 with User1 directly and select Read all SQL endpoint data.
- B. Assign User1 the Viewer role for Workspace1. Share Lakehouse1 with User1 and select Read all SQL endpoint data.
- C. Share Lakehouse1 with User1 directly and select Build reports on the default semantic model.
- D. Assign User1 the Member role for Workspace1. Share Lakehouse1 with User1 and select Read all SQL endpoint data.

Answer(s): A

QUESTION: 26

DRAG DROP (Drag and Drop is not supported)

You are implementing the following data entities in a Fabric environment:

1. Entity1: Available in a lakehouse and contains data that will be used as a core organization entity
2. Entity2: Available in a semantic model and contains data that meets organizational standards
3. Entity3: Available in a Microsoft Power BI report and contains data that is ready for sharing and reuse
4. Entity4: Available in a Power BI dashboard and contains approved data for executive-level decision making

Your company requires that specific governance processes be implemented for the data.

You need to apply endorsement badges to the entities based on each entity's use case.

Which badge should you apply to each entity? To answer, drag the appropriate badges the correct entities. Each badge may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

Note: Each correct selection is worth one point.

Select and Place:

Badges	Answer Area
⋮ Certified	Entity1:
⋮ Master data	Entity2:
⋮ Promoted	Entity3:
⋮ Cannot be endorsed	Entity4:

A. See Explanation section for answer.

Answer(s): A

Explanation:

Badges

☐ Certified

☐ Master data

☐ Promoted

☐ Cannot be endorsed

Answer Area

Entity1: ☐ Master data

Entity2: ☐ Certified

Entity3: ☐ Promoted

Entity4: ☐ Certified

Explanation:

Entity1: Master data

Entity1 is available in a lakehouse and contains data that will be used as a core organizational entity. This type of data, essential for the organization's operation, qualifies as Master data, which is foundational and authoritative within the business.

Entity2: Certified

Entity2 is available in a semantic model and contains data that meets organizational standards. Since it conforms to defined standards and has been validated, it should be Certified, signifying that it has passed the required data quality checks and is trusted for reporting and analysis.

Entity3: Promoted

Entity3 is available in a Microsoft Power BI report and contains data that is ready for sharing and reuse. Since the data is actively promoted for sharing and reuse within the organization, it receives the Promoted badge. This signifies that the data has been curated for wider consumption and distribution.

Entity4: Certified

Entity4 is available in a Power BI dashboard and contains approved data for executive-level decision making. This data is trusted and vetted for high-level decision-making and is thus Certified, indicating that it meets rigorous standards and is fit for executive-level use.

QUESTION: 27

HOTSPOT (Drag and Drop is not supported)

You have three users named User1, User2, and User3.

You have the Fabric workspaces shown in the following table.

Name	Workspace admin
Workspace1	User1
Workspace2	User2

You have a security group named Group1 that contains User1 and User3.

The Fabric admin creates the domains shown in the following table.

Name	Domain admin
Domain1	User1
Domain2	User2

User1 creates a new workspace named Workspace3.

You add Group1 to the default domain of Domain1.

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
User3 has Viewer role access to Workspace3.	☐	☐
User3 has Domain contributor access to Domain1.	☐	☐
User2 has Contributor role access to Workspace3.	☐	☐

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
User3 has Viewer role access to Workspace3.	☒	☐
User3 has Domain contributor access to Domain1.	☐	☒
User2 has Contributor role access to Workspace3.	☐	☒

Explanation:

User3 has Viewer role access to Workspace3 - Yes

Since User1 is the admin of Domain1 and Group1 (which contains User1 and User3) has been added to Domain1, User3 inherits the permissions based on their membership in Group1. By default, members of Domain1 will likely have the Viewer role in any new workspaces created within Domain1 unless specifically restricted.

User3 has Domain contributor access to Domain1 - No

Although User3 is part of Group1, which is added to Domain1, User1 is the admin of Domain1, and there is no direct assignment of Domain contributor access to User3. User3 inherits Viewer permissions for the workspaces under Domain1, but not Domain contributor permissions.

User2 has Contributor role access to Workspace3 - No

User2 is the admin of Domain2 and has no explicit permissions on Workspace3, which was created by User1 (the admin of Workspace1). Since User2 is not an admin of Domain1 (which governs Workspace3), they do not automatically have the Contributor role in Workspace3. Access to the workspace is governed by the workspace admin or the domain admin, but no such permission has been explicitly assigned to User2 for Workspace3.

QUESTION: 28

You have two Fabric workspaces named Workspace1 and Workspace2.

You have a Fabric deployment pipeline named deployPipeline1 that deploys items from Workspace1 to Workspace2. DeployPipeline1 contains all the items in Workspace1.

You recently modified the items in Workspaces1.

The workspaces currently contain the items shown in the following table.

Workspace	Items
Workspace1	Model1 Notebook1 Report1 Lakehouse1 Pipeline1
Workspace2	Model1 Notebook2 Report1 Lakehouse2

Items in Workspace1 that have the same name as items in Workspace2 are currently paired.

You need to ensure that the items in Workspace1 overwrite the corresponding items in Workspace2. The solution must minimize effort.

What should you do?

- A. Delete all the items in Workspace2, and then run deployPipeline1.
- B. Rename each item in Workspace2 to have the same name as the items in Workspace1.
- C. Back up the items in Workspace2, and then run deployPipeline1.
- D. Run deployPipeline1 without modifying the items in Workspace2.

Answer(s): D

Explanation:

When running a deployment pipeline in Fabric, if the items in Workspace1 are paired with the corresponding items in Workspace2 (based on the same name), the deployment pipeline will automatically overwrite the existing items in Workspace2 with the modified items from Workspace1. There's no need to delete, rename, or back up items manually unless you need to keep versions. By simply running deployPipeline1, the pipeline will handle overwriting the existing items in Workspace2 based on the pairing, ensuring the latest version of the items is deployed with minimal effort.

QUESTION: 29

You have a Fabric workspace named Workspace1 that contains a data pipeline named Pipeline1 and a lakehouse named Lakehouse1.

You have a deployment pipeline named deployPipeline1 that deploys Workspace1 to Workspace2.

You restructure Workspace1 by adding a folder named Folder1 and moving Pipeline1 to Folder1.

You use deployPipeline1 to deploy Workspace1 to Workspace2.

What occurs to Workspace2?

- A. Folder1 is created, Pipeline1 moves to Folder1, and Lakehouse1 is deployed.
- B. Only Pipeline1 and Lakehouse1 are deployed.
- C. Folder1 is created, and Pipeline1 and Lakehouse1 move to Folder1.
- D. Only Folder1 is created and Pipeline1 moves to Folder1.

Answer(s): A

Explanation:

When you restructure Workspace1 by adding a new folder (Folder1) and moving Pipeline1 into it, deployPipeline1 will deploy the entire structure of Workspace1 to Workspace2, preserving the changes made in Workspace1. This includes:

Folder1 will be created in Workspace2, mirroring the structure in Workspace1.

Pipeline1 will be moved into Folder1 in Workspace2, maintaining the same folder structure.

Lakehouse1 will be deployed to Workspace2 as it exists in Workspace1.

QUESTION: 30

DRAG DROP (Drag and Drop is not supported)

Your company has a team of developers. The team creates Python libraries of reusable code that is used to transform data.

You create a Fabric workspace name Workspace1 that will be used to develop extract, transform, and load (ETL) solutions by using notebooks.

You need to ensure that the libraries are available by default to new notebooks in Workspace1.

Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.

Select and Place:

Actions	Answer Area
⋮ Change the runtime version.	
⋮ Install the libraries.	
⋮ Create a pool.	
⋮ Create an environment.	
⋮ Set the default environment.	

A. See Explanation section for answer.

Answer(s): A

Explanation:

Actions
☐ Change the runtime version.

☐ Install the libraries.

☐ Create a pool.

☐ Create an environment.

☐ Set the default environment.
Answer Area
☐ Create an environment.

☐ Install the libraries.

☐ Set the default environment.
Explanation:

To ensure that your Python libraries are available by default to new notebooks in Workspace1, you need to follow these steps:

1. Create an environment: First, you need to create an environment where your libraries can be installed. This environment will be used by notebooks to access the required libraries.
2. Install the libraries: After creating the environment, you install the required Python libraries into that environment. These libraries can then be used by notebooks within Workspace1.
3. Set the default environment: Finally, to make sure that the libraries are available to new notebooks by default, you set this environment as the default environment for notebooks.

When you create a new notebook, it will automatically use the environment with the installed libraries.

QUESTION: 31

You have a Fabric workspace that contains a lakehouse and a notebook named Notebook1. Notebook1 reads data into a DataFrame from a table named Table1 and applies transformation logic. The data from the DataFrame is then written to a new Delta table named Table2 by using a merge operation.

You need to consolidate the underlying Parquet files in Table1.

Which command should you run?

- A. VACUUM
- B. BROADCAST
- C. OPTIMIZE
- D. CACHE

Answer(s): C**Explanation:**

To consolidate the underlying Parquet files in Table1 and improve query performance by optimizing the data layout, you should use the OPTIMIZE command in Delta Lake. The OPTIMIZE command coalesces smaller files into larger ones and reorganizes the data for more efficient reads. This is particularly useful when working with large datasets in Delta tables, as it helps reduce the number of files and improves performance for subsequent queries or operations like MERGE.

QUESTION: 32

You have five Fabric workspaces.

You are monitoring the execution of items by using Monitoring hub.

You need to identify in which workspace a specific item runs.

Which column should you view in Monitoring hub?

- A. Start time
- B. Capacity
- C. Activity name
- D. Submitter
- E. Item type
- F. Job type
- G. Location

Answer(s): G

Explanation:

To identify in which workspace a specific item runs in Monitoring hub, you should view the Location column.

This column indicates the workspace where the item is executed. Since you have multiple workspaces and need to track the execution of items across them, the Location column will show you the exact workspace associated with each item or job execution.

QUESTION: 33

You have a Fabric workspace that contains a warehouse named DW1. DW1 is loaded by using a notebook named Notebook1.

You need to identify which version of Delta was used when Notebook1 was executed.

What should you use?

- A. Real-Time hub
- B. OneLake data hub
- C. the Admin monitoring workspace
- D. Fabric Monitor
- E. the Microsoft Fabric Capacity Metrics app

Answer(s): D

QUESTION: 34

DRAG DROP (Drag and Drop is not supported)

You have a Fabric workspace that contains a warehouse named Warehouse1.

In Warehouse1, you create a table named DimCustomer by running the following statement.

```
CREATE TABLE dbo.DimCustomer (
    CustomerKey VARCHAR(255) NOT NULL,
    Name VARCHAR(255) NOT NULL,
    Email VARCHAR(255) NOT NULL
);
```

You need to set the Customerkey column as a primary key of the DimCustomer table.

Which three code segments should you run in sequence? To answer, move the appropriate code segments from the list of code segments to the answer area and arrange them in the correct order. Select and Place:

Code Segments

```

DROP CONSTRAINT PK_DimCustomer

ADD CONSTRAINT PK_DimCustomer PRIMARY KEY NONCLUSTERED
(CustomerKey)

NOT ENFORCED

ALTER TABLE dbo.DimCustomer

ADD CONSTRAINT PK_DimCustomer PRIMARY KEY CLUSTERED
(CustomerKey)

ENFORCED

```

Answer Area

A. See Explanation section for answer.

Answer(s): A

Explanation:

Code Segments

```

DROP CONSTRAINT PK_DimCustomer

ADD CONSTRAINT PK_DimCustomer PRIMARY KEY NONCLUSTERED
(CustomerKey)

NOT ENFORCED

ALTER TABLE dbo.DimCustomer

ADD CONSTRAINT PK_DimCustomer PRIMARY KEY CLUSTERED
(CustomerKey)

ENFORCED

```

Answer Area

```

ALTER TABLE dbo.DimCustomer

ADD CONSTRAINT PK_DimCustomer PRIMARY KEY CLUSTERED
(CustomerKey)

ENFORCED

```

Explanation:

To set the CustomerKey column as the primary key of the DimCustomer table, the steps need to follow the logical sequence of ensuring that the table structure can accommodate the primary key constraint.

1. ALTER TABLE dbo.DimCustomer

This command prepares the table for modification. It ensures the DimCustomer table is being updated to include the primary key.

2. ADD CONSTRAINT PK_DimCustomer PRIMARY KEY CLUSTERED (CustomerKey) This command creates a clustered primary key constraint on the CustomerKey column. A primary key uniquely identifies each record and enforces uniqueness.

3. ENFORCED

This ensures that the primary key constraint is enforced, meaning it will actively validate and enforce uniqueness and nullability on the CustomerKey column.

QUESTION: 35

You have a Fabric workspace that contains a semantic model named Model1.

You need to dynamically execute and monitor the refresh progress of Model1.

What should you use?

- A. dynamic management views in Microsoft SQL Server Management Studio (SSMS)
- B. Monitoring hub
- C. dynamic management views in Azure Data Studio
- D. a semantic link in a notebook

Answer(s): D

Explanation:

Semantic models in Microsoft Fabric are part of Power BI datasets and require refreshes to stay updated with the latest data.

Dynamically executing and monitoring the refresh progress requires a tool or approach that integrates with Fabric's capabilities for semantic models.

QUESTION: 36

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have a Fabric eventstream that loads data into a table named Bike_Location in a KQL database. The table contains the following columns:

BikepointID

Street

Neighbourhood

No_Bikes

No_Empty_Docks

Timestamp

You need to apply transformation and filter logic to prepare the data for consumption. The solution must return data for a neighbourhood named Sands End when No_Bikes is at least 15. The results must be ordered by No_Bikes in ascending order.

Solution: You use the following code segment:

```
bike_location
| filter Neighbourhood == "Sands End" and No_Bikes >= 15
| sort by No_Bikes
| project BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp
| project BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp
```

Does this meet the goal?

A. Yes

B. No

Answer(s): B

Explanation:

This code does not meet the goal because it uses sort by without specifying the order, which defaults to ascending, but explicitly mentioning asc improves clarity.

Correct code should look like:

```
bike_location
| filter Neighbourhood == "Sands End" and No_Bikes >= 15
| sort by No_Bikes asc
| project BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp
```

QUESTION: 37

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have a Fabric eventstream that loads data into a table named Bike_Location in a KQL database. The table contains the following columns:

BikepointID
Street
Neighbourhood
No_Bikes
No_Empty_Docks
Timestamp

You need to apply transformation and filter logic to prepare the data for consumption. The solution must return data for a neighbourhood named Sands End when No_Bikes is at least 15. The results must be ordered by No_Bikes in ascending order.

Solution: You use the following code segment:

```
bike_location
| filter Neighbourhood == "Sands End" and No_Bikes >= 15
| order by No_Bikes
| project BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp
```

Does this meet the goal?

- A. Yes
- B. No

Answer(s): B

Explanation:

This code does not meet the goal because it uses order by, which is not valid in KQL. The correct term in KQL is sort by.

Correct code should look like:

```
bike_location
| filter Neighbourhood == "Sands End" and No_Bikes >= 15
| sort by No_Bikes asc
| project BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp
```

QUESTION: 38

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have a Fabric eventstream that loads data into a table named Bike_Location in a KQL database. The table contains the following columns:

BikepointID
Street
Neighbourhood
No_Bikes
No_Empty_Docks
Timestamp

You need to apply transformation and filter logic to prepare the data for consumption. The solution must return data for a neighbourhood named Sands End when No_Bikes is at least 15. The results must be ordered by No_Bikes in ascending order.

Solution: You use the following code segment:

```
bike_location
| filter Neighbourhood == "Sands End" and No_Bikes >= 15
| sort by No_Bikes asc
| project BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp
```

Does this meet the goal?

- A. Yes
- B. No

Answer(s): A

Explanation:

Filter Condition: It correctly filters rows where Neighbourhood is "Sands End" and No_Bikes is greater than or equal to 15.

Sorting: The sorting is explicitly done by No_Bikes in ascending order using sort by No_Bikes asc.

Projection: It projects the required columns (BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp), which minimizes the data returned for consumption.

QUESTION: 39

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have a Fabric eventstream that loads data into a table named Bike_Location in a KQL database. The table contains the following columns:

```
BikepointID
Street
Neighbourhood
No_Bikes
No_Empty_Docks
Timestamp
```

You need to apply transformation and filter logic to prepare the data for consumption. The solution must return data for a neighbourhood named Sands End when No_Bikes is at least 15. The results must be ordered by No_Bikes in ascending order.

Solution: You use the following code segment:

```
SELECT BikepointID, Street, Neighbourhood, No_Bikes, No_Empty_Docks, Timestamp
FROM bike_location
WHERE neighbourhood = 'Sands End'
AND no_bikes >= 15
ORDER BY no_bikes
```

Does this meet the goal?

- A. Yes
- B. No

Answer(s): A

QUESTION: 40

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more

than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have an Azure key vault named KeyVault1 that contains secrets.

You have a Fabric workspace named Workspace1. Workspace contains a notebook named Notebook1 that performs the following tasks:

Loads stage data to the target tables in a lakehouse

Triggers the refresh of a semantic model

You plan to add functionality to Notebook1 that will use the Fabric API to monitor the semantic model refreshes.

You need to retrieve the registered application ID and secret from KeyVault1 to generate the authentication token.

Solution: You use the following code segment:

Use `notebookutils.credentials.getSecret` and specify the key vault URL and key vault secret.

Does this meet the goal?

A. Yes

B. No

Answer(s): A

Explanation:

Correct:

* You use the following code segment:

Use `notebookutils.credentials.getSecret` and specify the key vault URL and key vault secret.

Incorrect:

* You use the following code segment:

Use `notebookutils.credentials.putSecret` and specify the key vault URL and key vault secret.

* You use the following code segment:

Use `notebookutils.credentials.getSecret` and specify the key vault URL and the name of a linked service.

Note: Help on module `notebookutils.credentials` in `notebookutils`:

NAME

`notebookutils.credentials` - Utility for credentials operations in Fabric FUNCTIONS

`getSecret(akvName, secret) -> str`

Gets a secret from the given Azure Key Vault.

:param akvName: The name of the Azure Key Vault.

:param secret: The name of the secret.

:return: The secret value.

`notebookutils.credentials.getSecret('https://<name>.vault.azure.net/', 'secret name')`

QUESTION: 41

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have an Azure key vault named KeyVault1 that contains secrets.

You have a Fabric workspace named Workspace1. Workspace contains a notebook named Notebook1 that performs the following tasks:

Loads stage data to the target tables in a lakehouse

Triggers the refresh of a semantic model

You plan to add functionality to Notebook1 that will use the Fabric API to monitor the semantic model refreshes.

You need to retrieve the registered application ID and secret from KeyVault1 to generate the authentication token.

Solution: You use the following code segment:

Use `notebookutils.credentials.putSecret` and specify the key vault URL and key vault secret.
Does this meet the goal?

- A. Yes
- B. No

Answer(s): B

Explanation:

Correct:

* You use the following code segment:

Use `notebookutils.credentials.getSecret` and specify the key vault URL and key vault secret.

Incorrect:

* You use the following code segment:

Use `notebookutils.credentials.putSecret` and specify the key vault URL and key vault secret.

* You use the following code segment:

Use `notebookutils.credentials.getSecret` and specify the key vault URL and the name of a linked service.

Note: Help on module `notebookutils.credentials` in `notebookutils`:

NAME

`notebookutils.credentials` - Utility for credentials operations in Fabric FUNCTIONS

`getSecret(akvName, secret) -> str`

Gets a secret from the given Azure Key Vault.

:param akvName: The name of the Azure Key Vault.

:param secret: The name of the secret.

:return: The secret value.

`notebookutils.credentials.getSecret('https://<name>.vault.azure.net/', 'secret name')`

QUESTION: 42

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have an Azure key vault named KeyVault1 that contains secrets.

You have a Fabric workspace named Workspace1. Workspace contains a notebook named Notebook1 that performs the following tasks:

Loads stage data to the target tables in a lakehouse

Triggers the refresh of a semantic model

You plan to add functionality to Notebook1 that will use the Fabric API to monitor the semantic model refreshes.

You need to retrieve the registered application ID and secret from KeyVault1 to generate the authentication token.

Solution: You use the following code segment:

Use `notebookutils.credentials.getSecret` and specify the key vault URL and the name of a linked service.

Does this meet the goal?

- A. Yes
- B. No

Answer(s): B

Explanation:

Correct:

* You use the following code segment:

Use `notebookutils.credentials.getSecret` and specify the key vault URL and key vault secret.

Incorrect:

* You use the following code segment:

Use `notebookutils.credentials.putSecret` and specify the key vault URL and key vault secret.

* You use the following code segment:

Use `notebookutils.credentials.getSecret` and specify the key vault URL and the name of a linked service.

Note: Help on module `notebookutils.credentials` in `notebookutils`:

NAME

`notebookutils.credentials` - Utility for credentials operations in Fabric

FUNCTIONS

`getSecret(akvName, secret) -> str`

Gets a secret from the given Azure Key Vault.

:param akvName: The name of the Azure Key Vault.

:param secret: The name of the secret.

:return: The secret value.

`notebookutils.credentials.getSecret('https://<name>.vault.azure.net/', 'secret name')`

QUESTION: 43

DRAG DROP (Drag and Drop is not supported)

You have two Fabric notebooks named `Load_Salesperson` and `Load_Orders` that read data from Parquet files in a lakehouse. `Load_Salesperson` writes to a Delta table named `dim_salesperson`. `Load_Orders` writes to a Delta table named `fact_orders` and is dependent on the successful execution of `Load_Salesperson`.

You need to implement a pattern to dynamically execute `Load_Salesperson` and `Load_Orders` in the appropriate order by using a notebook.

How should you complete the code? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

Note: Each correct selection is worth one point.

Select and Place:

Values

Answer Area

```

DAG = {
  "  ": [
    {
      "name": "Load_Salesperson",
      "path": "Load_Salesperson",
      "timeoutPerCellInSeconds": 300,
    },
    {
      "name": "Load_Orders",
      "path": "Load_Orders",
      "timeoutPerCellInSeconds": 600,
      "  ": ["Load_Salesperson"]
    }
  ],
  "timeoutInSeconds": 43200
}

mssparkutils.notebook.  (DAG)

```

A. See Explanation section for answer.

Answer(s): A

Explanation:

Values

activities

broadcast

dependencies

execute

notebooks

runMultiple

Answer Area

```

DAG = {
  "activities": [
    {
      "name": "Load_Salesperson",
      "path": "Load_Salesperson",
      "timeoutPerCellInSeconds": 300,
    },
    {
      "name": "Load_Orders",
      "path": "Load_Orders",
      "timeoutPerCellInSeconds": 600,
      "dependencies": ["Load_Salesperson"]
    }
  ],
  "timeoutInSeconds": 43200
}

mssparkutils.notebook.runMultiple (DAG)

```

Box 1: activities

The following is an example of running notebooks with topological structure using `mssparkutils.notebook.runMultiple()`. Use this method to easily orchestrate notebooks through a code experience.

run multiple notebooks with parameters

```

DAG = {
  "activities": [
    {
      "name": "NotebookSimple", # activity name, must be unique
      "path": "NotebookSimple", # notebook path
      "timeoutPerCellInSeconds": 90, # max timeout for each cell, default to 90 seconds
      "args": {"p1": "changed value", "p2": 100}, # notebook parameters },
    {
      "name": "NotebookSimple2",
      "path": "NotebookSimple2",
      "timeoutPerCellInSeconds": 120,
      "args": {"p1": "changed value 2", "p2": 200}
    },
    {
      "name": "NotebookSimple2.2",
      "path": "NotebookSimple2",
      "timeoutPerCellInSeconds": 120,
      "args": {"p1": "changed value 3", "p2": 300},
      "retry": 1,
      "retryIntervallInSeconds": 10,

```



```
"dependencies": ["NotebookSimple"] # list of activity names that this activity depends on }
],
"timeoutInSeconds": 43200, # max timeout for the entire DAG, default to 12 hours "concurrency": 50 #
max number of notebooks to run concurrently, default to 50 }
mssparkutils.notebook.runMultiple(DAG, {"displayDAGViaGraphviz": False})
```

Box 2: dependencies

Box 3: runMultiple

Reference run multiple notebooks in parallel

This feature is in preview.

The method `mssparkutils.notebook.runMultiple()` allows you to run multiple notebooks in parallel or with a predefined topological structure. The API is using a multi-thread implementation mechanism within a spark session, which means the compute resources are shared by the reference notebook runs.

With `mssparkutils.notebook.runMultiple()`, you can:

- * Execute multiple notebooks simultaneously, without waiting for each one to finish.

- *-> Specify the dependencies and order of execution for your notebooks, using a simple JSON format.

- * Etc.

Here's a simple example of running a list of notebooks in parallel using this method:

```
mssparkutils.notebook.runMultiple(["NotebookSimple", "NotebookSimple2"])
```

QUESTION: 44

HOTSPOT (Drag and Drop is not supported)

You have a Fabric workspace named Workspace1 that contains a warehouse named Warehouse2.

A team of data analysts has Viewer role access to Workspace1.

You create a table by running the following statement.

```
CREATE TABLE [warehouse2].[dbo].[CreditCard]
(
    CreditCard varchar(20) NOT NULL
    ,CreditCardType varchar(10) NOT NULL)
GO
```

You need to ensure that the team can view only the first two characters and the last four characters of the CreditCard attribute.

How should you complete the statement? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

	▼
ALTER	
CREATE	
DEFAULT	
DROP	
EMAIL	
PARTIAL	
REPLACE	
UPDATE	

TABLE dbo.CreditCard

	▼
ALTER	
CREATE	
DEFAULT	
DROP	
EMAIL	
PARTIAL	
REPLACE	
UPDATE	

COLUMN [CreditCard]

ADD MASKED

WITH (FUNCTION = ')

	▼
ALTER	
CREATE	
DEFAULT	
DROP	
EMAIL	
PARTIAL	
REPLACE	
UPDATE	

(2, "XXXXXXXXXX", 4) ')

A. See Explanation section for answer.

Answer(s): A**Explanation:**

Answer Area

	TABLE dbo.CreditCard
ALTER	
CREATE	
DEFAULT	
DROP	
EMAIL	
PARTIAL	
REPLACE	
UPDATE	

	COLUMN [CreditCard]
ALTER	
CREATE	
DEFAULT	
DROP	
EMAIL	
PARTIAL	
REPLACE	
UPDATE	

ADD MASKED

WITH (FUNCTION = ' (2, "XXXXXXXXXX",4) ')

ALTER
CREATE
DEFAULT
DROP
EMAIL
PARTIAL
REPLACE
UPDATE

Box 1: ALTER

How to implement dynamic data masking in Fabric Data Warehouse

Configure dynamic data masking

Manage and modify dynamic data masking rules

To manage or modify existing dynamic data masking rules, create a new SQL script.

You can add a mask to an existing column, using the MASKED WITH FUNCTION clause:

ALTER TABLE dbo.EmployeeData

ALTER COLUMN [email] ADD MASKED WITH (FUNCTION = 'email()'); GO

Box 2: ALTER

Box 3: PARTIAL

As for the SSN column in the example below.

Note: In your SQL script, define dynamic data masking rules using the MASKED WITH FUNCTION clause. For example:

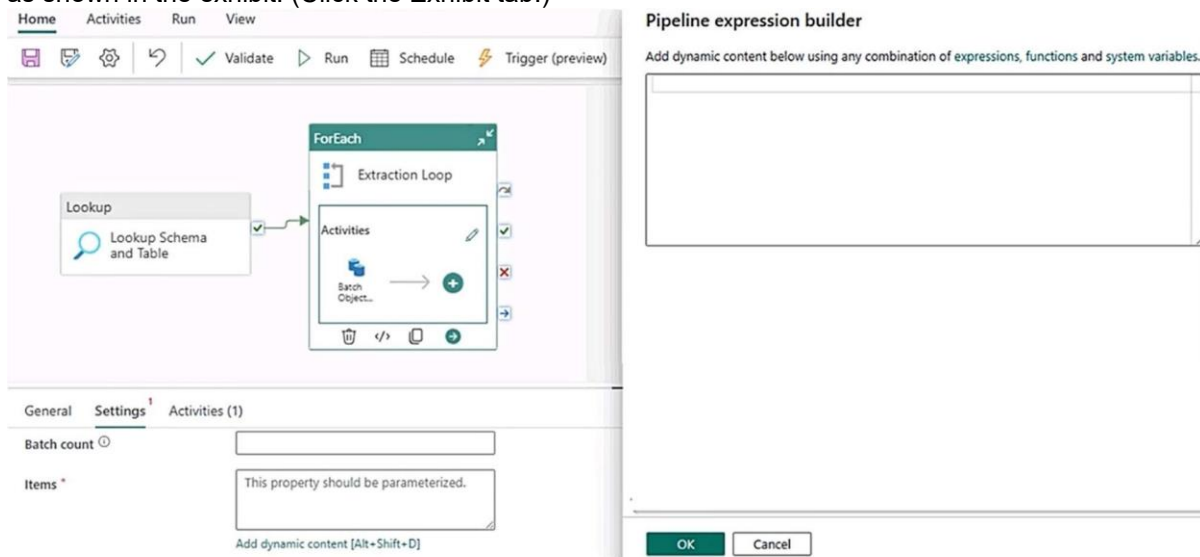
CREATE TABLE dbo.EmployeeData (

```
EmployeeID INT
,FirstName VARCHAR(50) MASKED WITH (FUNCTION = 'partial(1,"-",2)') NULL ,LastName
VARCHAR(50) MASKED WITH (FUNCTION = 'default()') NULL ,SSN CHAR(11) MASKED WITH
(FUNCTION = 'partial(0,"XXX-XX-",4)') NULL ,email VARCHAR(256) NULL
);
GO
INSERT INTO dbo.EmployeeData
VALUES (1, 'TestFirstName', 'TestLastName', '123-45-6789','email@youremail.com'); GO
INSERT INTO dbo.EmployeeData
VALUES (2, 'First_Name', 'Last_Name', '000-00-0000','email2@youremail2.com'); GO
The FirstName column shows only the first and last two characters of the string, with - in the middle.
The LastName column shows XXXX.
The SSN column shows XXX-XX- followed by the last four characters of the string.
```

QUESTION: 45

HOTSPOT (Drag and Drop is not supported)

You are building a data orchestration pattern by using a Fabric data pipeline named Dynamic Data Copy as shown in the exhibit. (Click the Exhibit tab.)



Dynamic Data Copy does NOT use parametrization.

You need to configure the ForEach activity to receive the list of tables to be copied.

How should you complete the pipeline expression? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

@activity('

Batch Object Copy
Dynamic Data Copy
Extraction Loop
Lookup Schema and Table

 ') .

output
output.count
output.pipelineReturnValue
output.value

A. See Explanation section for answer.

Answer(s): A**Explanation:****Answer Area**

```
@activity('
  Batch Object Copy
  Dynamic Data Copy
  Extraction Loop
  Lookup Schema and Table
').output
  output
  output.count
  output.pipelineReturnValue
  output.value
```

Box 1: Lookup Schema and Table

Copy multiple tables in bulk by using Azure Data Factory in the Azure portal

The ForEach activity iterates through a list of tables, and passes one by one to the Copy activity.

Example [replace LookupTableList with Lookup Schema and Table]:

In the Parameters section, click the input box under VALUE -> select the Add dynamic content below -> enter @activity('LookupTableList').output.value as table name value -> select Finish. You're setting the result list from the Lookup activity as an input to the second pipeline. The result list contains the list of tables whose data needs to be copied to the destination.

General Settings User properties				
Invoked pipeline *	IterateAndCopySQLTables		Open	+ New
Wait on completion	☐			
Parameters				
NAME	TYPE	VALUE	DEFAULT VALUE	
tableList	array	@activity('LookupTableList').output.value		

Box 2: output.value**QUESTION: 46**

HOTSPOT (Drag and Drop is not supported)

You have a Fabric workspace that contains a warehouse named Warehouse1. Warehouse1 contains a table named DimCustomers. DimCustomers contains the following columns:

CustomerName

CustomerID

BirthDate

EmailAddress

You need to configure security to meet the following requirements:

BirthDate in DimCustomer must be masked and display 1900-01-01.

EmailAddress in DimCustomer must be masked and display only the first leading character and the last five characters.

How should you complete the statement? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

```
ALTER TABLE DimCustomer
```

```
ALTER COLUMN BirthDate
```

```
ADD MASKED WITH (FUNCTION =
```

'default()'
'partial(1900-01-01)'
'random(1900-01-01, 1900-01-01)'

```
ALTER TABLE DimCustomer
```

```
ALTER COLUMN EmailAddress
```

```
ADD MASKED WITH (FUNCTION =
```

'default()'
'email()'
'partial(1,"@",5)'
'random(1,"@",5)'

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

```
ALTER TABLE DimCustomer
```

```
ALTER COLUMN BirthDate
```

```
ADD MASKED WITH (FUNCTION =
```

'default()'
'partial(1900-01-01)'
'random(1900-01-01, 1900-01-01)'

```
ALTER TABLE DimCustomer
```

```
ALTER COLUMN EmailAddress
```

```
ADD MASKED WITH (FUNCTION =
```

'default()'
'email()'
'partial(1,"@",5)'
'random(1,"@",5)'

Box 1: random(1900-01-01, 1900-01-01)

BirthDate in DimCustomer must be masked and display 1900-01-01.

Random

A random masking function for use on any numeric type to mask the original value with a random value within a specified range. Example definition syntax: Account_Number bigint MASKED WITH (FUNCTION = 'random ([start range], [end range])')

Example of alter syntax: ALTER COLUMN [Month] ADD MASKED WITH (FUNCTION = 'random(1, 12)')

Box 2: 'partial(1,"@",5)'

EmailAddress in DimCustomer must be masked and display only the first leading character and the last five characters.

Custom String Masking method that exposes the first and last letters and adds a custom padding string in the middle. prefix,[padding],suffix

If the original value is too short to complete the entire mask, part of the prefix or suffix isn't exposed.

Example definition syntax: FirstName varchar(100) MASKED WITH (FUNCTION = 'partial(prefix, [padding],suffix)') NULL

Example of alter syntax: ALTER COLUMN [Phone Number] ADD MASKED WITH (FUNCTION = 'partial(1,"XXXXXXX",0)')

This turns a phone number like 555.123.1234 into 5XXXXXXX.

Incorrect:

* 'default()'

Full masking according to the data types of the designated fields.

default() displays XXXX.

QUESTION: 47

You have a Fabric workspace named Workspace1 that contains the following items:

A Microsoft Power BI report named Report1

A Power BI dashboard named Dashboard1

A semantic model named Model1

A lakehouse name Lakehouse1

Your company requires that specific governance processes be implemented for the items.

Which items can you endorse in Fabric?

- A. Lakehouse1, Model1, and Dashboard1 only
- B. Lakehouse1, Model1, Report1 and Dashboard1
- C. Report1 and Dashboard1 only
- D. Model1, Report1, and Dashboard1 only
- E. Lakehouse1, Model1, and Report1 only

Answer(s): E

Explanation:

Not the Power BI dashboard Dashboard1.

Note: Endorsement is a Fabric and Power BI feature that makes it easier for users in an organization to find high-quality, trust-worthy content and data. Endorsed items are clearly labeled in the UI with a badge, and in some lists they are given precedence and are listed first.

Items can receive one of three endorsement badges - Promoted, Certified, or Master data:

Promoted: When an item has the Promoted badge, it means that the item creators think the item is ready for sharing and reuse. Any Fabric or Power BI item except Power BI dashboards can be promoted. Any user with write permissions on an item can promote it.

Certified: When an item has the Certified badge, it means that an organization-authorized reviewer has certified that the item meets the organization's quality standards, can be regarded as reliable and authoritative, and is ready for use across the organization. Any Fabric or Power BI item except Power BI dashboards can be certified. Any user can request certification for an item, but only users specified by a Fabric administrator can actually certify items.

QUESTION: 48

You have a Fabric workspace named Workspace1.

Your company acquires GitHub licenses.

You need to configure source control for Workspace1 to use GitHub. The solution must follow the principle of least privilege.

Which permissions do you require to ensure that you can commit code to GitHub?

- A. Actions (Read and write) and Contents (Read and write)
- B. Actions (Read and write) only
- C. Contents (Read and write) only
- D. Contents (Read) and Commit statuses (Read and write)

Answer(s): C

Explanation:

Microsoft Fabric Git integration

Git roles

The following table describes the Git permissions needed to perform various common operations:

If you're using a fine-grained access token, the following permissions are needed for common operations:

* Commit workspace changes to Git

Contents= Access: Read and write branch policy should allow direct commit

Note:

* Commit workspace changes to Git

All of the following roles:

Contributor in the workspace (WRITE permission on all items) Owner of the item (if the tenant switch blocks updates for nonowners) BUILD on external dependencies (where applicable)

QUESTION: 49

You have a Fabric workspace named Workspace1.

You plan to configure Git integration for Workspace1 by using an Azure DevOps Git repository.

An Azure DevOps admin creates the required artifacts to support the integration of Workspace1.

Which details do you require to perform the integration?

- A. the organization, project, Git repository, and branch
- B. the personal access token (PAT) for Git authentication and the Git repository URL
- C. the project, Git repository, branch, and Git folder
- D. the Git repository URL and the Git folder

Answer(s): A

Explanation:

Microsoft Fabric Git integration

Connect to your new Azure DevOps git repository, then click on the Connect and sync button:

 Connect Git repository and branch

Organization *

▼

Project *

▼

Git repository * ⓘ

▼

Branch * ⓘ

▼

Git folder ⓘ

QUESTION: 50

You have a Fabric workspace that contains a lakehouse and a semantic model named Model1. You use a notebook named Notebook1 to ingest and transform data from an external data source. You need to execute Notebook1 as part of a data pipeline named Pipeline1. The process must meet the following requirements:

- Run daily at 07:00 AM UTC.
- Attempt to retry Notebook1 twice if the notebook fails.
- After Notebook1 executes successfully, refresh Model1.

Which three actions should you perform? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. Place the Semantic model refresh activity after the Notebook activity and link the activities by using the On success condition.
- B. From the Schedule settings of Pipeline1, set the time zone to UTC.
- C. Set the Retry setting of the Notebook activity to 2.
- D. From the Schedule settings of Notebook1, set the time zone to UTC.
- E. Set the Retry setting of the Semantic model refresh activity to 2.
- F. Place the Semantic model refresh activity after the Notebook activity and link the activities by using an On completion condition.

Answer(s): A, B, C

Explanation:

- A (not F): Only execute the Semantic model refresh activity if Notebook1 is successful B (not D): The pipeline is to be scheduled, not the notebook activity.
- C (not E): Attempt to retry Notebook1 twice if the notebook fails.

QUESTION: 51

You have a Fabric workspace that contains a lakehouse named Lakehouse1.

You plan to create a data pipeline named Pipeline1 to ingest data into Lakehouse1. You will use a parameter named param1 to pass an external value into Pipeline1. The param1 parameter has a data type of int.

You need to ensure that the pipeline expression returns param1 as an int value.

How should you specify the parameter value?

- A. "@pipeline().parameters.param1"
- B. "@{pipeline().parameters.param1}"
- C. "@{pipeline().parameters.[param1]}"
- D. "@@{pipeline().parameters.param1}"

Answer(s): B

Explanation:

Parameters for Data Factory in Microsoft Fabric

Expressions can appear inside strings, using a feature called string interpolation where expressions are wrapped in @{ ... }. For example, the following string includes parameter values and literal string values: "First Name: @{pipeline().parameters.firstName} Last Name: @{pipeline().parameters.lastName}" Using string interpolation, the result is always a string. For example, if you defined myNumber as 42:

"@pipeline().parameters.myNumber"

Returns 42 as a number.

QUESTION: 52

You have a Fabric workspace named Workspace1 that contains a lakehouse named Lakehouse1.

Workspace1 contains the following items:

A Dataflow Gen2 dataflow that copies data from an on-premises Microsoft SQL Server database to Lakehouse1

A notebook that transforms files and loads the data to Lakehouse1

A data pipeline that loads a CSV file to Lakehouse1

You need to develop an orchestration solution in Fabric that will load each item one after the other. The solution must be scheduled to run every 15 minutes.

Which type of item should you use?

- A. notebook
- B. warehouse
- C. Dataflow Gen2 dataflow
- D. data pipeline

Answer(s): D

Explanation:

Ingest data by using a data pipeline, dataflow, or notebook

Fabric provides three basic ways to retrieve data from existing storage and systems: data pipelines, dataflows, and notebooks. Each item uses a different user approach and targets different types of users.

Data pipelines

A data pipeline is an item from the Data Factory experience that acts as an orchestration component. It can run other items and services and be scheduled to run at specific times.

Incorrect:

Not A: Notebooks are items that are used primarily for creating Apache Spark jobs and machine learning experiments.

QUESTION: 53

You are building a Fabric notebook named MasterNotebook1 in a workspace. MasterNotebook1 contains the following code.

```

DAG = {
    "activities": [
        {
            "name": "execute_notebook_1",
            "path": "notebook_01",
            "timeoutPerCellInSeconds": 600,
            "args": {
                "input_value": "999"
            },
            "retry": 1,
            "retryIntervalInSeconds": 30
        },
        {
            "name": "execute_notebook_2",
            "path": "notebook_02",
            "timeoutPerCellInSeconds": 400,
            "args": {
                "input_value": "888"
            },
            "retry": 1,
            "retryIntervalInSeconds": 30
        },
        {
            "name": "execute_notebook_3",
            "path": "notebook_03",
            "timeoutPerCellInSeconds": 600,
            "args": {
                "input_value": "777"
            },
            "retry": 1,
            "retryIntervalInSeconds": 30
        },
        {
            "name": "execute_notebook_3",
            "path": "notebook_03",
            "timeoutPerCellInSeconds": 600,
            "args": {
                "input_value": "777"
            },
            "retry": 1,
            "retryIntervalInSeconds": 30
        }
    ],
    "timeoutInSeconds": 43200,
    "concurrency": 0
}

```

You need to ensure that the notebooks are executed in the following sequence:

1. Notebook_03
2. Notebook_01
3. Notebook_02

Which two actions should you perform? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. Move the declaration of Notebook_02 to the bottom of the Directed Acyclic Graph (DAG) definition.
- B. Add dependencies to the execution of Notebook_03.
- C. Split the Directed Acyclic Graph (DAG) definition into three separate definitions.
- D. Add dependencies to the execution of Notebook_02.
- E. Change the concurrency to 3.
- F. Move the declaration of Notebook_03 to the top of the Directed Acyclic Graph (DAG) definition.

Answer(s): D, F

Explanation:

D: Add dependencies to the execution of the last notebook, which is Notebook_02. It depends on Notebook_03 and Notebook_01.

Incorrect:

Not B: Notebook_03 should be executed first, it does not depend on any other job.

Not E: We do not want them to run in parallel. We want to run in a sequence.

Note: Reference run multiple notebooks in parallel

The method `mssparkutils.notebook.runMultiple()` allows you to run multiple notebooks in parallel or with a predefined topological structure. The API is using a multi-thread implementation mechanism within a spark session, which means the compute resources are shared by the reference notebook runs.

With `mssparkutils.notebook.runMultiple()`, you can:

- * Execute multiple notebooks simultaneously, without waiting for each one to finish.

- *-> Specify the dependencies and order of execution for your notebooks, using a simple JSON format.

- * Etc.

The following is an example of running notebooks with topological structure using `mssparkutils.notebook.runMultiple()`. Use this method to easily orchestrate notebooks through a code experience.

run multiple notebooks with parameters

```
DAG = {
  "activities": [
    {
      "name": "NotebookSimple", # activity name, must be unique
      "path": "NotebookSimple", # notebook path
      "timeoutPerCellInSeconds": 90, # max timeout for each cell, default to 90 seconds
      "args": {"p1": "changed value", "p2": 100}, # notebook parameters },
    {
      "name": "NotebookSimple2",
      "path": "NotebookSimple2",
      "timeoutPerCellInSeconds": 120,
      "args": {"p1": "changed value 2", "p2": 200}
    },
    {
      "name": "NotebookSimple2.2",
      "path": "NotebookSimple2",
      "timeoutPerCellInSeconds": 120,
      "args": {"p1": "changed value 3", "p2": 300},
      "retry": 1,
      "retryIntervallInSeconds": 10,
      "dependencies": ["NotebookSimple"] # list of activity names that this activity depends on }
  ]
}
```

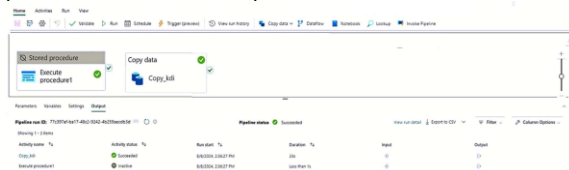
```

],
"timeoutInSeconds": 43200, # max timeout for the entire DAG, default to 12 hours "concurrency": 50 #
max number of notebooks to run concurrently, default to 50 }
mssparkutils.notebook.runMultiple(DAG, {"displayDAGViaGraphviz": False})

```

QUESTION: 54

You have a Fabric workspace that contains a data pipeline named Pipeline1 as shown in the exhibit. (Click the Exhibit tab.)



What will occur the next time Pipeline1 runs?

- A. Copy_kdi will run first, and then Execute procedure1 will run.
- B. Execute procedure1 will run first, and then Copy_kdi will run.
- C. Execute procedure1 will run and Copy_kdi will be skipped.
- D. Copy_kdi will run and Execute procedure1 will be skipped.
- E. Both activities will run simultaneously.
- F. Both activities will be skipped.

Answer(s): F

QUESTION: 55

HOTSPOT (Drag and Drop is not supported)

You have a Fabric workspace that contains a warehouse named Warehouse1. Warehouse1 contains a table named Customer. Customer contains the following data.

CustomerID	FirstName	LastName	Phone	CreditCard
1	John	Doe	555-123-4567	1234567812345670
2	Jane	Smith	555-987-6543	8765432187654320
3	Michael	Johnson	555-555-5555	1234987654321230
4	Emily	Davis	555-222-3333	4321123456789870
5	David	Brown	555-444-5555	5678123498761230

You have an internal Microsoft Entra user named User1 that has an email address of user1@contoso.com.

You need to provide User1 with access to the Customer table. The solution must prevent User1 from accessing the CreditCard column.

How should you complete the statement? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

GRANT ON

ALTER
EXECUTE
READ
SELECT
VIEW

Customers(CustomerID, FirstName, LastName, Phone)

TO

User1
[User1]
[user1@contoso.com]

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

GRANT ON

ALTER
EXECUTE
READ
SELECT
VIEW

Customers(CustomerID, FirstName, LastName, Phone)

TO

User1
[User1]
[user1@contoso.com]

Box 1: SELECT

Implement column-level security in Fabric data warehousing

Column-level security (CLS) in Microsoft Fabric allows you to control access to columns in a table based on specific grants on these tables.

Define column-level access for tables

1. Identify user or roles and the data tables you want to secure with column-level security.
2. Implement column-level security with the GRANT T-SQL statement and a column list. For simplicity of management, assigning permissions to roles is preferred to using individuals.

-- Grant select to subset of columns of a table

GRANT SELECT ON YourSchema.YourTable

(Column1, Column2, Column3, Column4, Column5)

TO [SomeGroup];

Replace YourSchema with the name of your schema and YourTable with the name of your target table.

3. Replace SomeGroup with the name of your User/Group.

4. Replace the comma-delimited columns list with the columns you want to give the role access to.

5. Repeat these steps to grant specific column access for other tables if needed.

Box 2: [User1]

QUESTION: 56

You have a Fabric deployment pipeline that uses three workspaces named Dev, Test, and Prod.

You need to deploy an Eventhouse as part of the deployment process.

What should you use to add the Eventhouse to the deployment process?

- A. an Azure DevOps pipeline
- B. an eventstream
- C. GitHub Actions

Answer(s): A

Explanation:

Correct:

- * an Azure DevOps pipeline

Incorrect:

- * a deployment pipeline
- * an eventstream
- * GitHub Actions

Reference:

<https://oulasvirta.com/data/2024/10/15/the-enterprise-fabric-challenge-devops-for-microsoft-fabric.html>

QUESTION: 57

You have a Fabric warehouse named DW1. DW1 contains a table that stores sales data and is used by multiple sales representatives.

You plan to implement row-level security (RLS).

You need to ensure that the sales representatives can see only their respective data.

Which warehouse object do you require to implement RLS?

- A. TRIGGER
- B. SCHEMA
- C. FUNCTION
- D. DATABASE ROLE

Answer(s): C

Explanation:

Correct:

- * Function

Restrict access to certain rows to certain users

Implement RLS by using the CREATE SECURITY POLICY Transact-SQL statement, and predicates created as inline table-valued functions.

* SECURITY POLICY

Incorrect:

* CONSTRAINT

* DATABASE ROLE

* SCHEMA

* STORED PROCEDURE

* TABLE

* TRIGGER

Reference:

<https://learn.microsoft.com/en-us/fabric/data-warehouse/row-level-security>

QUESTION: 58

You have a Fabric warehouse named DW1. DW1 contains a table that stores sales data and is used by multiple sales representatives.

You plan to implement row-level security (RLS).

You need to ensure that the sales representatives can see only their respective data.

Which warehouse object do you require to implement RLS?

A. SECURITY POLICY

B. TABLE

C. TRIGGER

D. STORED PROCEDURE

Answer(s): A

Explanation:

Correct:

* Function

Restrict access to certain rows to certain users

Implement RLS by using the CREATE SECURITY POLICY Transact-SQL statement, and predicates created as inline table-valued functions.

* SECURITY POLICY

Incorrect:

* CONSTRAINT

* DATABASE ROLE

* SCHEMA

* STORED PROCEDURE

* TABLE

* TRIGGER

Reference:

<https://learn.microsoft.com/en-us/fabric/data-warehouse/row-level-security>

QUESTION: 59

You have a Fabric F32 capacity that contains a workspace. The workspace contains a warehouse named DW1 that is modelled by using MD5 hash surrogate keys.

DW1 contains a single fact table that has grown from 200 million rows to 500 million rows during the past year.

You have Microsoft Power BI reports that are based on Direct Lake. The reports show year-over-year values.

Users report that the performance of some of the reports has degraded over time and some visuals show errors.

You need to resolve the performance issues. The solution must meet the following requirements:
 Provide the best query performance.
 Minimize operational costs.
 Which should you do?

- A. Create views.
- B. Modify the surrogate keys to use a different data type.
- C. Change the MD5 hash to SHA256.
- D. Increase the capacity.
- E. Disable V-Order on the warehouse.

Answer(s): D

Explanation:

Correct:

* Increase the capacity.

Direct Lake in Fabric F32 Capacity supports tables up to 300 million rows. F64 can handle 500 million rows.

Incorrect:

* Change the MD5 hash to SHA256.

* Create views.

* Disable V-Order on the warehouse.

* Enable V-Order.

* Modify the surrogate keys to use a different data type.

Reference:

<https://community.fabric.microsoft.com/t5/Power-BI-Community-Blog/Fabric-Capacity-Scaling-and-Power-BI-What-happens-when-Power-BI/ba-p/4403792>

QUESTION: 60

You have a Fabric workspace named Workspace1 that contains a warehouse named Warehouse1.

You plan to deploy Warehouse 1 to a new workspace named Workspace2.

As part of the deployment process, you need to verify whether Warehouse1 contains invalid references. The solution must minimize development effort and provide detailed information about the invalid references.

What should you use?

- A. a dbt project
- B. a deployment pipeline
- C. a Python script
- D. a database project

Answer(s): B

Explanation:

Correct:

* a deployment pipeline

Incorrect:

* a database project

* a dbt project

* a Python script

* a T-SQL script

Note:

A deployment pipeline in Fabric allows you to deploy assets like warehouses, datasets, and reports between different workspaces (such as from Workspace1 to Workspace2). One of the key features of a deployment pipeline is the ability to check for invalid references before deployment. This can help identify

issues with assets, such as broken links or dependencies, ensuring the deployment is successful without introducing errors. This is the most efficient way to verify references and manage the deployment with minimal development effort.

Reference:

<https://learn.microsoft.com/en-us/fabric/cicd/deployment-pipelines/understand-the-deployment-process>

Exam Topic: Ingest and transform data (Testlet 1)

Case Study:

This is a case study. Case studies are not timed separately. You can use as much exam time as you would like to complete each case. However, there may be additional case studies and sections on this exam.

You must manage your time to ensure that you are able to complete all questions included on this exam in the time provided.

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At the end of this case study, a review screen will appear. This screen allows you to review your answers and to make changes before you move to the next section of the exam. After you begin a new section, you cannot return to this section.

To start the case study

To display the first question in this case study, click the Next button. Use the buttons in the left pane to explore the content of the case study before you answer the questions. Clicking these buttons displays information such as business requirements, existing environment, and problem statements. If the case study has an All Information tab, note that the information displayed is identical to the information displayed on the subsequent tabs.

When you are ready to answer a question, click the Question button to return to the question.

Overview:

Litware, Inc. is a publishing company that has an online bookstore and several retail bookstores worldwide.

Litware also manages an online advertising business for the authors it represents.

Existing Environment. Fabric Environment

Litware has a Fabric workspace named Workspace1. High concurrency is enabled for Workspace1.

The company has a data engineering team that uses Python for data processing.

Existing Environment. Data Processing

The retail bookstores send sales data at the end of each business day, while the online bookstore constantly provides logs and sales data to a central enterprise resource planning (ERP) system. Litware implements a medallion architecture by using the following three layers: bronze, silver, and gold. The sales data is ingested from the ERP system as Parquet files that land in the Files folder in a lakehouse.

Notebooks are used to transform the files in a Delta table for the bronze and silver layers. The gold layer is in a warehouse that has V-Order disabled.

Litware has image files of book covers in Azure Blob Storage. The files are loaded into the Files folder.

Existing Environment. Sales Data

Month-end sales data is processed on the first calendar day of each month. Data that is older than one month never changes.

In the source system, the sales data refreshes every six hours starting at midnight each day. The sales data is captured in a Dataflow Gen2 dataflow. When the dataflow runs, new and historical data is captured. The dataflow captures the following fields of the source:

Sales Date

Author

Price

Units

SKU

A table named AuthorSales stores the sales data that relates to each author. The table contains a column named AuthorEmail. Authors authenticate to a guest Fabric tenant by using their email address.

Existing Environment. Security Groups

Litware has the following security groups:

Sales

Fabric Admins

Streaming Admins

Existing Environment. Performance Issues

Business users perform ad-hoc queries against the warehouse. The business users indicate that reports against the warehouse sometimes run for two hours and fail to load as expected. Upon further investigation, the data engineering team receives the following error message when the reports fail to load: "The SQL query failed while running."

The data engineering team wants to debug the issue and find queries that cause more than one failure. When the authors have new book releases, there is often an increase in sales activity. This increase slows the data ingestion process.

The company's sales team reports that during the last month, the sales data has NOT been up-to-date when they arrive at work in the morning.

Requirements. Planned Changes

Litware recently signed a contract to receive book reviews. The provider of the reviews exposes the data in Amazon Simple Storage Service (Amazon S3) buckets.

Litware plans to manage Search Engine Optimization (SEO) for the authors. The SEO data will be streamed from a REST API.

Requirements. Version Control

Litware plans to implement a version control solution in Fabric that will use GitHub integration and follow the principle of least privilege.

Requirements. Governance Requirements

To control data platform costs, the data platform must use only Fabric services and items. Additional Azure resources must NOT be provisioned.

Requirements. Data Requirements

Litware identifies the following data requirements:

Process the SEO data in near-real-time (NRT).

Make the book reviews available in the lakehouse without making a copy of the data.

When a new book cover image arrives in the Files folder, process the image as soon as possible.

QUESTION: 61

You need to create a workflow for the new book cover images.

Which two components should you include in the workflow? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. an activator item
- B. a data pipeline
- C. a blob storage action
- D. a time-based schedule
- E. a streaming dataflow
- F. a notebook that uses Apache Spark Structured Streaming

Answer(s): C, E

Explanation:

Correct:

- * a streaming dataflow
- * a blob storage action

Incorrect:

- * a data pipeline
- * a notebook that uses Apache Spark Structured Streaming
- * a reflex item
- * a time-based schedule
- * an activator item

Note:

- * a streaming dataflow

Streaming dataflows allow authors to connect to, ingest, mash up, model, and build reports based on streaming in near real-time data directly in the Power BI service.

- * a blob storage action

Azure Blob storage is Microsoft's object storage solution for the cloud. Blob storage is optimized for storing massive amounts of unstructured data. Unstructured data is data that doesn't adhere to a particular data model or definition, such as text or binary data.

You can use Azure Blobs as a streaming or reference input. Streaming blobs are checked every second for updates. Unlike a streaming blob, a reference blob is only loaded at the beginning of the refresh. It's static data that isn't expected to change, and the recommended limit for static data is 50 MB or less.

Connect to Azure Blob Storage from workflows in Azure Logic Apps You can access your Azure Blob Storage account and container from a workflow in Azure Logic Apps using the Azure Blob Storage connector. This connector provides triggers and actions that your workflow can use for blob operations. You can then create automated workflows that run when triggered by events in your storage container or in other systems, and run actions to work with data in your storage container. For example, you can access and manage files stored as blobs in your Azure storage account.

Scenario:

Requirements. Data Requirements

When a new book cover image arrives in the Files folder, process the image as soon as possible.

Existing Environment. Data Processing

The sales data is ingested from the ERP system as Parquet files that land in the Files folder in a lakehouse.

Notebooks are used to transform the files in a Delta table for the bronze and silver layers. The gold layer is in a warehouse that has V-Order disabled.

*-> Litware has image files of book covers in Azure Blob Storage. The files are loaded into the Files folder.

QUESTION: 62

You need to create a workflow for the new book cover images.

Which two components should you include in the workflow? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. a time-based schedule
- B. a streaming dataflow
- C. a blob storage action
- D. a data pipeline
- E. a notebook that uses Apache Spark Structured Streaming
- F. a reflex item

Answer(s): B, C

Explanation:

BC: Streaming dataflows

Streaming dataflows allow authors to connect to, ingest, mash up, model, and build reports based on streaming in near real-time data directly in the Power BI service.

Blob storage

Azure Blob storage is Microsoft's object storage solution for the cloud. Blob storage is optimized for storing massive amounts of unstructured data. Unstructured data is data that doesn't adhere to a particular data model or definition, such as text or binary data.

You can use Azure Blobs as a streaming or reference input. Streaming blobs are checked every second for updates. Unlike a streaming blob, a reference blob is only loaded at the beginning of the refresh. It's static data that isn't expected to change, and the recommended limit for static data is 50 MB or less.

C: Connect to Azure Blob Storage from workflows in Azure Logic Apps You can access your Azure Blob Storage account and container from a workflow in Azure Logic Apps using the Azure Blob Storage connector. This connector provides triggers and actions that your workflow can use for blob operations. You can then create automated workflows that run when triggered by events in your storage container or in other systems, and run actions to work with data in your storage container. For example, you can access and manage files stored as blobs in your Azure storage account.

Scenario:

Requirements. Data Requirements

When a new book cover image arrives in the Files folder, process the image as soon as possible.

Existing Environment. Data Processing

The sales data is ingested from the ERP system as Parquet files that land in the Files folder in a lakehouse.

Notebooks are used to transform the files in a Delta table for the bronze and silver layers. The gold layer is in a warehouse that has V-Order disabled.

*-> Litware has image files of book covers in Azure Blob Storage. The files are loaded into the Files folder.

QUESTION: 63

What should you recommend that the data engineering team use to ingest the SEO data?

- A. a streaming dataflow
- B. a streaming dataset
- C. a notebook that uses Apache Spark Structured Streaming
- D. an eventstream

Answer(s): D

Explanation:

Microsoft Fabric Real-Time Intelligence, Eventstream REST API The Microsoft Fabric REST APIs enable you to automate Fabric procedures and processes, helping your organization complete tasks more efficiently and accurately. By automating these workflows, you can reduce errors, improve productivity, and achieve cost savings across your operations.

In Fabric, an item represents a set of capabilities within a specific experience. For example, Eventstream

is an item under the Real-time Intelligence experience. Each item in Fabric is defined by an item definition--an object that outlines the structure, format, and key components that make up the item.

Scenario:

Requirements. Planned Changes

Litware plans to manage Search Engine Optimization (SEO) for the authors. The SEO data will be streamed from a REST API.

Requirements. Data Requirements

Litware identifies the following data requirements:

*-> Process the SEO data in near-real-time (NRT).

QUESTION: 64

You need to implement the solution for the book reviews.
Which should you do?

- A. Create a Dataflow Gen2 dataflow.
- B. Create a shortcut.
- C. Enable external data sharing.
- D. Create a data pipeline.

Answer(s): B

Explanation:

The requirement specifies that Litware plans to make the book reviews available in the lakehouse without making a copy of the data. In this case, creating a shortcut in Fabric is the most appropriate solution. A shortcut is a reference to the external data, and it allows Litware to access the book reviews stored in Amazon S3 without duplicating the data into the lakehouse.

QUESTION: 65

You need to resolve the sales data issue. The solution must minimize the amount of data transferred.
What should you do?

- A. Split the dataflow into two dataflows.
- B. Configure scheduled refresh for the dataflow.
- C. Configure incremental refresh for the dataflow. Set Store rows from the past to 1 Month.
- D. Configure incremental refresh for the dataflow. Set Refresh rows from the past to 1 Year.
- E. Configure incremental refresh for the dataflow. Set Refresh rows from the past to 1 Month.

Answer(s): E

Explanation:

The sales data issue can be resolved by configuring incremental refresh for the dataflow. Incremental refresh allows for only the new or changed data to be processed, minimizing the amount of data transferred and improving performance.

The solution specifies that data older than one month never changes, so setting the refresh period to 1 Month is appropriate. This ensures that only the most recent month of data will be refreshed, reducing unnecessary data transfers.

Exam Topic: Ingest and transform data (Testlet 2)

Case Study:

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When you are ready to answer a question, click the Question button to return to the question.

Overview:

- Company Overview

Contoso, Ltd. is an online retail company that wants to modernize its analytics platform by moving to Fabric.

The company plans to begin using Fabric for marketing analytics.

Overview:

- IT Structure

The company's IT department has a team of data analysts and a team of data engineers that use analytics systems.

The data engineers perform the ingestion, transformation, and loading of data. They prefer to use Python or SQL to transform the data.

The data analysts query data and create semantic models and reports. They are qualified to write queries in Power Query and T-SQL.

Existing Environment. Fabric

Contoso has an F64 capacity named Cap1. All Fabric users are allowed to create items.

Contoso has two workspaces named WorkspaceA and WorkspaceB that currently use Pro license mode.

Existing Environment. Source Systems

Contoso has a point of sale (POS) system named POS1 that uses an instance of SQL Server on Azure Virtual Machines in the same Microsoft Entra tenant as Fabric. The host virtual machine is on a private virtual network that has public access blocked. POS1 contains all the sales transactions that were processed on the company's website.

The company has a software as a service (SaaS) online marketing app named MAR1. MAR1 has seven entities. The entities contain data that relates to email open rates and interaction rates, as well as website interactions. The data can be exported from MAR1 by calling REST APIs. Each entity has a different endpoint.

Contoso has been using MAR1 for one year. Data from prior years is stored in Parquet files in an Amazon Simple Storage Service (Amazon S3) bucket. There are 12 files that range in size from 300 MB to 900 MB and relate to email interactions.

Existing Environment. Product Data

POS1 contains a product list and related data. The data comes from the following three tables:

Products

ProductCategories

ProductSubcategories

In the data, products are related to product subcategories, and subcategories are related to product categories.

Existing Environment. Azure

Contoso has a Microsoft Entra tenant that has the following mail-enabled security groups:

DataAnalysts: Contains the data analysts

DataEngineers: Contains the data engineers

Contoso has an Azure subscription.

The company has an existing Azure DevOps organization and creates a new project for repositories that relate to Fabric.

Existing Environment. User Problems

The VP of marketing at Contoso requires analysis on the effectiveness of different types of email content. It typically takes a week to manually compile and analyze the data. Contoso wants to reduce the time to less than one day by using Fabric.

The data engineering team has successfully exported data from MAR1. The team experiences transient connectivity errors, which causes the data exports to fail.

Requirements. Planned Changes

Contoso plans to create the following two lakehouses:

Lakehouse1: Will store both raw and cleansed data from the sources

Lakehouse2: Will serve data in a dimensional model to users for analytical queries

Additional items will be added to facilitate data ingestion and transformation.

Contoso plans to use Azure Repos for source control in Fabric.

Requirements. Technical Requirements

The new lakehouses must follow a medallion architecture by using the following three layers: bronze, silver, and gold. There will be extensive data cleansing required to populate the MAR1 data in the silver layer, including deduplication, the handling of missing values, and the standardizing of capitalization. Each layer must be fully populated before moving on to the next layer. If any step in populating the lakehouses fails, an email must be sent to the data engineers.

Data imports must run simultaneously, when possible.

The use of email data from the Amazon S3 bucket must meet the following requirements:

Minimize egress costs associated with cross-cloud data access.

Prevent saving a copy of the raw data in the lakehouses.

Items that relate to data ingestion must meet the following requirements:

The items must be source controlled alongside other workspace items.

Ingested data must land in the bronze layer of Lakehouse1 in the Delta format.

No changes other than changes to the file formats must be implemented before the data lands in the bronze layer.

Development effort must be minimized and a built-in connection must be used to import the source data.

In the event of a connectivity error, the ingestion processes must attempt the connection again.

Lakehouses, data pipelines, and notebooks must be stored in WorkspaceA. Semantic models, reports, and dataflows must be stored in WorkspaceB.

Once a week, old files that are no longer referenced by a Delta table log must be removed.

Requirements. Data Transformation

In the POS1 product data, ProductID values are unique. The product dimension in the gold layer must include only active products from product list. Active products are identified by an IsActive value of 1.

Some product categories and subcategories are NOT assigned to any product. They are NOT analytically relevant and must be omitted from the product dimension in the gold layer.

Requirements. Data Security

Security in Fabric must meet the following requirements:

The data engineers must have read and write access to all the lakehouses, including the underlying files.
The data analysts must only have read access to the Delta tables in the gold layer.
The data analysts must NOT have access to the data in the bronze and silver layers.
The data engineers must be able to commit changes to source control in WorkspaceA.

QUESTION: 66

HOTSPOT (Drag and Drop is not supported)

You need to recommend a method to populate the POS1 data to the lakehouse medallion layers.
What should you recommend for each layer? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Bronze layer:

A Dataflow Gen2 dataflow
A notebook
A pipeline Copy activity
A pipeline stored procedure

Silver layer:

A Dataflow Gen2 dataflow
A notebook
A pipeline Copy activity
A pipeline stored procedure

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

Bronze layer:

Silver layer:

Explanation:

Bronze Layer: A pipeline Copy activity

The bronze layer is used to store raw, unprocessed data. The requirements specify that no transformations should be applied before landing the data in this layer. Using a pipeline Copy activity ensures minimal development effort, built-in connectors, and the ability to ingest the data directly into the Delta format in the bronze layer.

Silver Layer: A notebook

The silver layer involves extensive data cleansing (deduplication, handling missing values, and standardizing capitalization). A notebook provides the flexibility to implement complex transformations and is well-suited for this task.

QUESTION: 67

You need to ensure that usage of the data in the Amazon S3 bucket meets the technical requirements. What should you do?

- A. Create a workspace identity and enable high concurrency for the notebooks.
- B. Create a shortcut and ensure that caching is disabled for the workspace.
- C. Create a workspace identity and use the identity in a data pipeline.
- D. Create a shortcut and ensure that caching is enabled for the workspace.

Answer(s): D

QUESTION: 68

HOTSPOT (Drag and Drop is not supported)

You need to create the product dimension.

How should you complete the Apache Spark SQL code? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

```
SELECT ProductID, ProductNumber, ProductName, ModelName, SubCategoryName, CategoryName
```

```
FROM ContosoLake.Products p
```

ContosoLake.ProductSubCategories s ON p.SubCategoryID = s.SubCategoryID

FULL JOIN
INNER JOIN
LEFT ANTI JOIN
LEFT OUTER JOIN
OUTER JOIN

ContosoLake.ProductCategories c ON c.CategoryID = s.CategoryID

FULL JOIN
INNER JOIN
LEFT ANTI JOIN
LEFT OUTER JOIN
OUTER JOIN

WHERE

CategoryID = 1;
CategoryName is not null;
IsActive = 1;
IsActive is not null;
ProductNumber is not null;
SubCategoryID = 1;
SubCategoryName is not null;

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

```
SELECT ProductID, ProductNumber, ProductName, ModelName, SubCategoryName, CategoryName
```

```
FROM ContosoLake.Products p
```

FULL JOIN
 INNER JOIN
 LEFT ANTI JOIN
 LEFT OUTER JOIN
 OUTER JOIN

FULL JOIN
 INNER JOIN
 LEFT ANTI JOIN
 LEFT OUTER JOIN
 OUTER JOIN

WHERE

 CategoryID = 1;
 CategoryName is not null;
 IsActive = 1;
 IsActive is not null;
 ProductNumber is not null;
 SubCategoryID = 1;
 SubCategoryName is not null;

Explanation:

Join between Products and ProductSubCategories:

Use an INNER JOIN.

The goal is to include only products that are assigned to a subcategory. An INNER JOIN ensures that only matching records (i.e., products with a valid subcategory) are included.

Join between ProductSubCategories and ProductCategories:

Use an INNER JOIN.

Similar to the above logic, we want to include only subcategories assigned to a valid product category. An INNER JOIN ensures this condition is met.

WHERE Clause

Condition: IsActive = 1

Only active products (where IsActive equals 1) should be included in the gold layer. This filters out inactive products.

QUESTION: 69

You need to populate the MAR1 data in the bronze layer.

Which two types of activities should you include in the pipeline? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. ForEach
- B. Copy data
- C. WebHook
- D. Stored procedure

Answer(s): A, B

Explanation:

MAR1 has seven entities, each accessible via a different API endpoint. A ForEach activity is required to

iterate over these endpoints to fetch data from each one. It enables dynamic execution of API calls for each entity.

The Copy data activity is the primary mechanism to extract data from REST APIs and load it into the bronze layer in Delta format. It supports native connectors for REST APIs and Delta, minimizing development effort.

QUESTION: 70

You need to recommend a solution to resolve the MAR1 connectivity issues. The solution must minimize development effort.

What should you recommend?

- A. Add a ForEach activity to the data pipeline.
- B. Configure retries for the Copy data activity.
- C. Call a notebook from the data pipeline.
- D. Configure Fault tolerance for the Copy data activity.

Answer(s): B

Explanation:

Retry Feature in Azure Data Factory

Enhancing Data Pipeline Reliability with Retry Feature in Azure Data Factory Understanding the Retry Feature

The Retry feature in ADF's Copy activity allows users to automatically retry failed operations in the event of transient failures. These failures could be caused by factors such as network disruptions, Java Native Interface errors, or temporary service unavailability. By configuring the retry settings, users can specify the number of retry attempts and the interval between retries, providing flexibility to tailor the retry behavior according to specific requirements.

Scenario:

Requirements. Technical Requirements

Items that relate to data ingestion must meet the following requirements:

*-> In the event of a connectivity error, the ingestion processes must attempt the connection again.

* Etc.

Note: Existing environment:

The company has a software as a service (SaaS) online marketing app named MAR1. MAR1 has seven entities. The entities contain data that relates to email open rates and interaction rates, as well as website interactions. The data can be exported from MAR1 by calling REST APIs. Each entity has a different endpoint.

Contoso has been using MAR1 for one year.

Requirements. Technical Requirements

The new lakehouses must follow a medallion architecture by using the following three layers: bronze, silver, and gold. There will be extensive data cleansing required to populate the MAR1 data in the silver layer, including deduplication, the handling of missing values, and the standardizing of capitalization.

QUESTION: 71

You need to recommend a solution for handling old files. The solution must meet the technical requirements.

What should you include in the recommendation?

- A. a data pipeline that includes a Copy data activity
- B. a data pipeline that includes a Delete data activity
- C. a notebook that runs the VACUUM command
- D. a notebook that runs the OPTIMIZE command

Answer(s): C

Explanation:

Scenario:

Technical requirements include:

Once a week, old files that are no longer referenced by a Delta table log must be removed.

Note: Remove unused data files with vacuum

Predictive optimization automatically runs VACUUM on Unity Catalog managed tables. Databricks recommends enabling predictive optimizations for all Unity Catalog managed tables to simplify data maintenance and reduce storage costs.

You can remove data files no longer referenced by a Delta table that are older than the retention threshold by running the VACUUM command on the table. Running VACUUM regularly is important for cost and compliance because of the following considerations:

Deleting unused data files reduces cloud storage costs.

Data files removed by VACUUM might contain records that have been modified or deleted. Permanently removing these files from cloud storage ensures these records are no longer accessible.

Exam Topic: Ingest and transform data (Question Set 3)

QUESTION: 72

HOTSPOT (Drag and Drop is not supported)

You have a Fabric workspace that contains a warehouse named Warehouse1. Warehouse1 contains the following tables and columns.

Table name	Column name	Data type
Employee	EmployeeID	Int
Employee	EmployeeName	Varchar(128)
Employee	EmployeePosition	Varchar(64)
Contract	EmployeeID	Int
Contract	ContractType	Varchar(64)
Contract	StartDate	Datetime2
Contract	EndDate	Datetime2

You need to denormalize the tables and include the ContractType and StartDate columns in the Employee table. The solution must meet the following requirements:

Ensure that the StartDate column is of the date data type.

Ensure that all the rows from the Employee table are preserved and include any matching rows from the Contract table.

Ensure that the result set displays the total number of employees per contract type for all the contract types that have more than two employees.

How should you complete the statement? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

A. See Explanation section for answer.

Answer(s): A

Explanation:

Explanation:

CAST

The requirement specifies that the StartDate column must be of the date data type. Using CAST (c.StartDate AS date) converts the StartDate from its current datetime2 data type to date.

LEFT OUTER JOIN

The requirement states that all rows from the Employee table must be preserved, even if there are no matching rows in the Contract table. A LEFT OUTER JOIN ensures this by including all rows from the Employee table and matching rows from the Contract table.

HAVING

The final condition filters the result set to include only contract types with more than two employees. The HAVING clause is used to filter groups (after applying GROUP BY) based on an aggregate function, in this case, COUNT(DISTINCT EmployeeID) > 2.

```
WITH result AS (
    SELECT
        e.EmployeeID,
        e.EmployeeName,
        e.EmployeePosition,
        c.ContractType,
        CAST(c.StartDate AS date) AS StartDate
    FROM Employee AS e
    LEFT OUTER JOIN Contract AS c ON c.EmployeeID = e.EmployeeID
)
SELECT
    COUNT(DISTINCT EmployeeID) AS TotalEmployees,
    ContractType
FROM result
GROUP BY ContractType
HAVING COUNT(DISTINCT EmployeeID) > 2;
```

QUESTION: 73

HOTSPOT (Drag and Drop is not supported)

You have an Azure Event Hubs data source that contains weather data.

You ingest the data from the data source by using an eventstream named Eventstream1. Eventstream1 uses a lakehouse as the destination.

You need to batch ingest only rows from the data source where the City attribute has a value of Kansas.

The filter must be added before the destination. The solution must minimize development effort.

What should you use for the data processor and filtering? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Data processor:

A data pipeline
A Dataflow Gen2 dataflow
An eventstream with a custom endpoint
An eventstream with an external data source

Filtering:

A Filter activity in a data pipeline
A filter in a Dataflow Gen2 dataflow
A KQL statement
An eventstream processor

A. See Explanation section for answer.

Answer(s): A**Explanation:****Answer Area**

Data processor:

A data pipeline
A Dataflow Gen2 dataflow
An eventstream with a custom endpoint
An eventstream with an external data source

Filtering:

A Filter activity in a data pipeline
A filter in a Dataflow Gen2 dataflow
A KQL statement
An eventstream processor

Explanation:

Data Processor - A Dataflow Gen2 dataflow

A Dataflow Gen2 dataflow is the best choice here because it is specifically designed for high-performance transformations and can be used for batch and stream processing. In this case, you need to filter the data (based on the City attribute having a value of Kansas) before it is written to the destination. Dataflow Gen2 offers a simple, no-code approach for transformations and is fully integrated with Microsoft Fabric, which minimizes development effort.

Filtering - A filter in a Dataflow Gen2 dataflow

Since the task is to filter rows based on the City attribute, the best approach is to apply the filter directly within the Dataflow Gen2 dataflow. You can define this filter with minimal code and no external dependencies. Dataflow Gen2 allows for efficient and scalable filtering on large data volumes before the data is ingested into the destination (lakehouse), which minimizes development effort.

QUESTION: 74

You have a Fabric workspace that contains an eventstream named Eventstream1. Eventstream1 processes data from a thermal sensor by using event stream processing, and then stores the data in a lakehouse.

You need to modify Eventstream1 to include the standard deviation of the temperature.

Which transform operator should you include in the Eventstream1 logic?

- A. Expand
- B. Group by
- C. Union
- D. Aggregate

Answer(s): B

QUESTION: 75

You have an Azure event hub. Each event contains the following fields:

BikepointID

Street

Neighbourhood

Latitude

Longitude

No_Bikes

No_Empty_Docks

You need to ingest the events. The solution must only retain events that have a Neighbourhood value of Chelsea, and then store the retained events in a Fabric lakehouse.

What should you use?

- A. a KQL queryset
- B. an eventstream
- C. a streaming dataset
- D. Apache Spark Structured Streaming

Answer(s): B

Explanation:

An eventstream is the best solution for ingesting data from Azure Event Hub into Fabric, while applying filtering logic such as retaining only the events that have a Neighbourhood value of "Chelsea."

Eventstreams in Microsoft Fabric are designed for handling real-time data streams and can apply transformation logic directly on incoming events. In this case, the eventstream can filter events based on the Neighbourhood field before storing the retained events in a Fabric lakehouse.

Eventstreams are well-suited for stream processing, such as this case where you need to filter out only specific data (events with a Neighbourhood of "Chelsea") before storing it in the lakehouse.

QUESTION: 76

HOTSPOT (Drag and Drop is not supported)

You are building a data loading pattern for Fabric notebook workloads.

You have the following code segment:

```
def loading_pattern_sample(df_source):
    try:
        deltaTable = DeltaTable.forName(spark, target_table)
    except Exception:
        try:
            df_source.write.format('delta').mode('overwrite').saveAsTable(f"{target_table}")
        except Exception as e:
            print(f'Load for table {target_table} failed with error: {str(e)}')
            raise
        return

    try:
        change_detection_columns = [col for col in df_source.columns if col not in candidate_key]

        match_condition = ' AND '.join([f'target.{col} = source.{col}' for col in candidate_key])
        update_condition = ' OR '.join([f'target.{col} != source.{col}' for col in change_detection_columns])

        update_expr = {col: f'source.{col}' for col in df_source.columns}

        merge_operation = deltaTable.alias('target').merge(
            source=df_source.alias('source'),
            condition=match_condition
        ).whenMatchedUpdate(
            condition=update_condition,
            set=update_expr
        ).whenNotMatchedInsertAll()

        merge_operation.execute()
    except Exception as e:
        print(f'Insert operation for table {target_table} failed with error: {str(e)}')
    return
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
The target table will always be overwritten.	☐	☐
The merge operation will always run.	☐	☐
The loading pattern supports both full and incremental loading requirements.	☐	☐

A. See Explanation section for answer.

Answer(s): A

Explanation:**Answer Area****Statements****Yes****No**

The target table will always be overwritten.

☐☒

The merge operation will always run.

☐☒

The loading pattern supports both full and incremental loading requirements.

☒☐**Explanation:**

The target table will always be overwritten: No

The code first checks if the target Delta table exists using `DeltaTable.forName`. If it does not exist, the table is created, and data is written using `df_source.write.format('delta').mode('overwrite')`. However, if the table exists, a merge operation is executed instead of overwriting the table. Therefore, the target table is not always overwritten.

The merge operation will always run: No

The merge operation only runs if the target Delta table already exists. If the table does not exist, the data is written directly without performing a merge. Therefore, the merge operation does not always run.

The loading pattern supports both full and incremental loading requirements: Yes The loading pattern supports full loads by overwriting the table when it does not exist and incremental loads through the merge operation. The merge operation detects changes (via `update_condition`) and inserts or updates records accordingly. This supports both full and incremental loading.

QUESTION: 77

HOTSPOT (Drag and Drop is not supported)

You have a Fabric workspace that contains two lakehouses named Lakehouse1 and Lakehouse2.

Lakehouse1 contains staging data in a Delta table named Orderlines. Lakehouse2 contains a Type 2 slowly changing dimension (SCD) dimension table named Dim_Customer.

You need to build a query that will combine data from Orderlines and Dim_Customer to create a new fact table named Fact_Orders. The new table must meet the following requirements:

Enable the analysis of customer orders based on historical attributes.

Enable the analysis of customer orders based on the current attributes.

How should you complete the statement? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

```

SELECT
    OrderLineID order_line_id
    ,OrderDate order_date
    ,c.customer_key
    ,c.customer_id
    ,Quantity order_quantity
    ,UnitPrice unit_price
    ,TaxRate tax_rate
FROM
    Lakehouse1.orderlines o
INNER JOIN
    Lakehouse2.dim_customer c
    ON o.customerid = c.customer_id

```

AND

c.is_current = 1
o.OrderDate > c.valid_to_datetime
o.OrderDate >= c.valid_from_datetime

AND

c.is_current = 1
o.OrderDate < c.valid_to_datetime
o.OrderDate <= c.valid_from_datetime

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

```

SELECT
  OrderLineID order_line_id
  ,OrderDate order_date
  ,c.customer_key
  ,c.customer_id
  ,Quantity order_quantity
  ,UnitPrice unit_price
  ,TaxRate tax_rate
FROM
  Lakehouse1.orderlines o
INNER JOIN
  Lakehouse2.dim_customer c
  ON o.customerid = c.customer_id

```

AND

AND

Explanation:

Historical Analysis: The query must link each order to the correct historical attributes from the Dim_Customer table. This is achieved by ensuring the OrderDate falls within the valid date range (valid_from_datetime and valid_to_datetime) of the customer record.

Current Analysis: The query must also allow analysis of orders based on the current attributes, which can be achieved by filtering where c.is_current = 1.

```

SELECT
    o.OrderLineID AS order_line_id,
    o.OrderDate AS order_date,
    c.customer_key,
    c.customer_id,
    o.Quantity AS order_quantity,
    o.UnitPrice AS unit_price,
    o.TaxRate AS tax_rate
FROM
    Lakehouse1.orderlines o
INNER JOIN
    Lakehouse2.dim_customer c
ON
    o.customerid = c.customer_id
AND
    o.OrderDate >= c.valid_from_datetime
AND
    o.OrderDate <= c.valid_to_datetime;

```

A . OrderDate >= c.valid_from_datetime

To enable historical analysis, the OrderDate should fall after or on the valid_from_datetime. This ensures that only orders placed within the valid time period of the customer record are linked.

B . OrderDate <= c.valid_to_datetime

For historical analysis, the OrderDate should also fall before or on the valid_to_datetime. This ensures orders are matched with the correct historical customer attributes.

QUESTION: 78

You have a Fabric workspace that contains a lakehouse named Lakehouse1.

In an external data source, you have data files that are 500 GB each. A new file is added every day.

You need to ingest the data into Lakehouse1 without applying any transformations. The solution must meet the following requirements

Trigger the process when a new file is added.

Provide the highest throughput.

Which type of item should you use to ingest the data?

- A. Eventstream
- B. Dataflow Gen2
- C. Streaming dataset
- D. Data pipeline

Answer(s): D

QUESTION: 79

You have a Fabric workspace that contains a lakehouse named Lakehouse1.

In an external data source, you have data files that are 500 GB each. A new file is added every day. You need to ingest the data into Lakehouse1 without applying any transformations. The solution must meet the following requirements

Trigger the process when a new file is added.

Provide the highest throughput.

Which type of item should you use to ingest the data?

- A. Data pipeline
- B. Environment
- C. KQL queryset
- D. Dataflow Gen2

Answer(s): A

Explanation:

To efficiently ingest large data files (500 GB each) into Lakehouse1 with high throughput and trigger the process when a new file is added, a Data pipeline is the most suitable solution. Data pipelines in Fabric are ideal for orchestrating data movement and can be configured to automatically trigger based on file arrivals or other events. This solution meets both requirements: ingesting the data without transformations (since you just need to copy the data) and triggering the process when new files are added.

QUESTION: 80

You have a Fabric workspace that contains an eventhouse and a KQL database named Database1.

Database1 has the following:

A table named Table1

A table named Table2

An update policy named Policy1

Policy1 sends data from Table1 to Table2.

The following is a sample of the data in Table2.

Timestamp (datetime)	DeviceId (guid)	StreamData (dynamic)
2024-05-18 12:45:17.16524	81416f30- 60a2-4e75- 9b19- 2a84ea059735	[{ "index": 0, "eventid": "719afca0- be30-4559-bb5e- 59feade642f6" }]
2024-05-18 12:45:21.76423	bb664e1e- 02aa-4e17- 8c8a- 116cd4458d52	[{ "index": 0, "eventid": "782222b2- fbc0-43c0-82d6-ecd49a99dbf5" }]
2024-05-18 12:45:23.98642	717bfe7d- 0e5d-498f- 9f21- e60aaf258056	[{ "index": 0, "eventid": "d5730286- 0da4-41f8-8e59-f75e209310a9" }]

Recently, the following actions were performed on Table1:

An additional element named temperature was added to the StreamData column.

The data type of the Timestamp column was changed to date.

The data type of the DeviceId column was changed to string.

You plan to load additional records to Table2.

Which two records will load from Table1 to Table2? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

A.

Timestamp (datetime)	DeviceId (guid)	StreamData (dynamic)
2024-05-18	81416f30- 60a2-4e75- 9b19- 2a84ea059735	[{ "index": 40, "eventid": "729afca2-be30-4559-bb5e-59feade642f3", "temperature": 32 }]

B.

Timestamp (datetime)	DeviceId (guid)	StreamData (dynamic)
2024-05-21	81416f30	[{ "index": 0, "eventId": "719afca0-be30-4559-bb5e-5werade642f6", "temperature": 27 }]

C.

Timestamp (datetime)	DeviceId (guid)	StreamData (dynamic)
2024-05-23	81416f3060a24e759b192a84ea05973532dhdyte3	[{ "index": 0, "eventId": "719afca0-be30-4559-bb5e-59feade642f6" }]

D.

Timestamp (datetime)	DeviceId (guid)	StreamData (dynamic)
2024-05-24	81416f30-60a2-4e75-9b19-2a84ea059735	[{ "index": 0, "eventId": "719afca0-be30-4559-bb5e-59feade642f6" }]

Answer(s): B, D**Explanation:**

Changes to Table1 Structure:

StreamData column: An additional temperature element was added.

Timestamp column: Data type changed from datetime to date.

DeviceId column: Data type changed from guid to string.

Impact of Changes:

Only records that comply with Table2's structure will load.

Records that deviate from Table2's column data types or structure will be rejected.

Record B:

Timestamp: Matches Table2 (datetime format).

DeviceId: Matches Table2 (guid format).

StreamData: Contains only the index and eventId, which matches Table2.

Accepted because it fully matches Table2's structure and data types.

Record D:

Timestamp: Matches Table2 (datetime format).

DeviceId: Matches Table2 (guid format).

StreamData: Matches Table2's structure.

Accepted because it fully matches Table2's structure and data types.

QUESTION: 81

HOTSPOT (Drag and Drop is not supported)

You have a Fabric workspace.

You are debugging a statement and discover the following issues:

Sometimes, the statement fails to return all the expected rows.

The PurchaseDate output column is NOT in the expected format of mmm dd, yy.

You need to resolve the issues. The solution must ensure that the data types of the results are retained.

The results can contain blank cells.

How should you complete the statement? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

SELECT

item_id as ItemId

	▼
,convert(varchar(20), item_name)	
,convert(varchar(max), item_name)	
,try_cast(item_name as varchar(20))	

as ItemName

,item_description as ItemDescription

	▼
,convert(varchar, purchase_date, 7)	
,convert(varchar, purchase_date, 109)	
,convert(varchar, purchase_date, 112)	

as PurchaseDate

FROM

Table1

WHERE

item_type = @itemtype_parameter

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

SELECT

item_id as ItemId

	▼
,convert(varchar(20), item_name)	
,convert(varchar(max), item_name)	
,try_cast(item_name as varchar(20))	

as ItemName

,item_description as ItemDescription

	▼
,convert(varchar, purchase_date, 7)	
,convert(varchar, purchase_date, 109)	
,convert(varchar, purchase_date, 112)	

as PurchaseDate

FROM

Table1

WHERE

item_type = @itemtype_parameter

```

SELECT
    item_id AS ItemId,
    try_cast(item_name as varchar(20)) AS ItemName,
    item_description AS ItemDescription,
    convert(varchar, purchase_date, 7) AS PurchaseDate
FROM
    Table1
WHERE
    item_type = @itemtype_parameter;

```

ItemName - try_cast(item_name as varchar(20))

try_cast attempts to convert the item_name column to varchar(20) and returns NULL if the conversion fails instead of raising an error. This ensures that the query does not fail and returns all rows, even if some conversions are unsuccessful.

PurchaseDate - convert(varchar, purchase_date, 7)

The style value 7 in CONVERT outputs dates in the format mmm dd, yy (e.g., "Aug 25, 21"). This matches the required format.

QUESTION: 82

You are developing a data pipeline named Pipeline1.

You need to add a Copy data activity that will copy data from a Snowflake data source to a Fabric warehouse.

Which option from the Settings tab of the Copy data activity must you configure?

- A. Degree of copy parallelism
- B. Fault tolerance
- C. Enable staging
- D. Enable logging

Answer(s): C

Explanation:

When using the Copy data activity in a data pipeline to move data from Snowflake to a Fabric warehouse, the process often involves intermediate staging to handle data efficiently, especially for large datasets or cross- cloud data transfers.

Staging involves temporarily storing data in an intermediate location (e.g., Blob storage or Azure Data Lake) before loading it into the target destination.

For cross-cloud data transfers (e.g., from Snowflake to Fabric), enabling staging ensures data is processed and stored temporarily in an efficient format for transfer.

Staging is especially useful when dealing with large datasets, ensuring the process is optimized and avoids memory limitations.

QUESTION: 83

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have a KQL database that contains two tables named Stream and Reference. Stream contains streaming data in the following format.

Column name	Data type
Timestamp	Datetime
GeoLocation	Dynamic
Temperature	Decimal
DeviceId	Int

Reference contains reference data in the following format.

Column name	Data type
DeviceId	Int
DeviceName	String

Both tables contain millions of rows.

You have the following KQL queryset.

```

01 Stream
02 | extend lat = todecimal(GeoLocation.Latitude), long = todecimal(GeoLocation.Longitude)
03 | join kind=inner Reference on DeviceId
04 | project Timestamp, lat, long, Temperature, DeviceName
05 | filter Temperature >= 10
06 | render scatterchart with (kind = map)

```

You need to reduce how long it takes to run the KQL queryset.

Solution: You change the join type to kind=outer.

Does this meet the goal?

A. Yes

B. No

Answer(s): B

Explanation:

An outer join will include unmatched rows from both tables, increasing the dataset size and processing time. It does not improve query performance.

QUESTION: 84

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

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Reference contains reference data in the following format.

Column name	Data type
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03 | join kind=inner Reference on DeviceId
04 | project Timestamp, lat, long, Temperature, DeviceName
05 | filter Temperature >= 10
06 | render scatterchart with (kind = map)
```

You need to reduce how long it takes to run the KQL queryset.

Solution: You change project to extend.

Does this meet the goal?

A. Yes

B. No

Answer(s): B

Explanation:

Using extend retains all columns in the table, potentially increasing the size of the output unnecessarily. project is more efficient because it selects only the required columns.

QUESTION: 85

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have a KQL database that contains two tables named Stream and Reference. Stream contains streaming data in the following format.

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Timestamp	Datetime
GeoLocation	Dynamic
Temperature	Decimal
DeviceId	Int

Reference contains reference data in the following format.

Column name	Data type
DeviceId	Int
DeviceName	String

Both tables contain millions of rows.

You have the following KQL queryset.

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03 | join kind=inner Reference on DeviceId
04 | project Timestamp, lat, long, Temperature, DeviceName
05 | filter Temperature >= 10
06 | render scatterchart with (kind = map)
```

You need to reduce how long it takes to run the KQL queryset.

Solution: You move the filter to line 02.

Does this meet the goal?

- A. Yes
- B. No

Answer(s): A

Explanation:

Moving the filter to line 02: Filtering the Stream table before performing the join operation reduces the number of rows that need to be processed during the join. This is an effective optimization technique for queries involving large datasets.

QUESTION: 86

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have a KQL database that contains two tables named Stream and Reference. Stream contains streaming data in the following format.

Column name	Data type
Timestamp	Datetime
GeoLocation	Dynamic
Temperature	Decimal
DeviceId	Int

Reference contains reference data in the following format.

Column name	Data type
DeviceId	Int
DeviceName	String

Both tables contain millions of rows.

You have the following KQL queryset.

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01 Stream
02 | extend lat = todecimal(GeoLocation.Latitude), long = todecimal(GeoLocation.Longitude)
03 | join kind=inner Reference on DeviceId
04 | project Timestamp, lat, long, Temperature, DeviceName
05 | filter Temperature >= 10
06 | render scatterchart with (kind = map)
```

You need to reduce how long it takes to run the KQL queryset.

Solution: You add the make_list() function to the output columns.

Does this meet the goal?

A. Yes

B. No

Answer(s): B

Explanation:

Adding an aggregation like make_list() would require additional processing and memory, which could make the query slower.

QUESTION: 87

HOTSPOT (Drag and Drop is not supported)

You have a Fabric warehouse named DW1 that contains four staging tables named ProductCategory, ProductSubcategory, Product, and SalesOrder. ProductCategory, ProductSubcategory, and Product are used often in analytical queries.

You need to implement a star schema for DW1. The solution must minimize development effort.

Which design approach should you use? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

ProductCategory, ProductSubcategory
and Product must be:

Added to the model as individual tables
Denormalized by being added to the SalesOrder table
Denormalized into a single product dimension table

The joining key must be:

The product name and the date
The unique system generated identifier
The product category name

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

ProductCategory, ProductSubcategory
and Product must be:

Added to the model as individual tables
Denormalized by being added to the SalesOrder table
Denormalized into a single product dimension table

The joining key must be:

The product name and the date
The unique system generated identifier
The product category name

Box 1: Denormalized into a single product dimension table

ProductCategory, ProductSubCategory, and Product must be:

SalesOrder is the Fact table.

Denormalize the Product dimensional table.

Note: Fact tables and dimension tables

A star schema is used to denormalize business data into dimensions (like time and product) and facts (like transactions in amounts and quantities).

A star schema has a single fact table in the center, containing business "facts" (like transaction amounts and quantities). The fact table connects to multiple other dimension tables along "dimensions" like time, or product. Star schemas enable users to slice and dice the data however they see fit, typically by joining two or more fact tables and dimension tables together.

Denormalized data

Star schemas denormalize the data, which means adding redundant columns to some dimension tables to make querying and working with the data faster and easier. The purpose is to trade some redundancy (duplication of data) in the data model for increased query speed, by avoiding computationally expensive join operations.

In this model, the fact table is normalized but the dimensions tables are not. That is, data from the fact table exists only on the fact table, but dimensional tables may hold redundant data.

Box 2: The product name and the date

The joining key must be:

QUESTION: 88

HOTSPOT (Drag and Drop is not supported)

Your company has three newly created data engineering teams named Team1, Team2, and Team3 that plan to use Fabric. The teams have the following personas:

Team1 consists of members who currently use Microsoft Power BI. The team wants to transform data by using a low-code approach.

Team2 consists of members that have a background in Python programming. The team wants to use PySpark code to transform data.

Team3 consists of members who currently use Azure Data Factory. The team wants to move data between source and sink environments by using the least amount of effort.

You need to recommend tools for the teams based on their current personas.

What should you recommend for each team? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Team1:	Data pipelines Notebooks Dataflow Gen2 dataflow
Team2:	Data pipelines Notebooks Dataflow Gen2 dataflow
Team3:	Data pipelines Notebooks Dataflow Gen2 dataflow

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

Team1:

Team2:

Team3:

Box 1: Dataflow Gen2 dataflow

Team1 consists of members who currently use Microsoft Power BI. The team wants to transform data by using by a low-code approach.

Dataflow Gen2 is the new generation of dataflows. The new generation of dataflows resides alongside the Power BI Dataflow (Gen1) and brings new features and improved experiences.

Working with Dataflow Gen2 feels like coming home. We kept the full Power Query experience you're used to in Power BI dataflows.

Box 2: Notebooks

Team2 consists of members that have a background in Python programming. The team wants to use PySpark code to transform data.

A Microsoft Fabric notebook is a primary code item for developing Apache Spark jobs and machine learning experiments. It's a web-based interactive surface used by data scientists and data engineers to write code benefiting from rich visualizations and Markdown text.

Fabric notebooks currently support four Apache Spark languages:

PySpark (Python)

Spark (Scala)

Spark SQL

SparkR

Box 3: Data pipelines

Team3 consists of members who currently use Azure Data Factory. The team wants to move data between source and sink environments by using the least amount of effort.

With Data Factory, you can use the Copy Activity in a data pipeline to move data from both on-premises and cloud source data stores to a centralization data store in the cloud for further analysis. For example, you can collect data in Azure Data Lake Storage and transform the data later by using an Azure Data Lake Analytics compute service. You can also collect data in Azure Blob storage and transform it later by using an Azure HDInsight Hadoop cluster.

QUESTION: 89

HOTSPOT (Drag and Drop is not supported)

You plan to process the following three datasets by using Fabric:

Dataset1: This dataset will be added to Fabric and will have a unique primary key between the source

and the destination. The unique primary key will be an integer and will start from 1 and have an increment of 1. Dataset2: This dataset contains semi-structured data that uses bulk data transfer. The dataset must be handled in one process between the source and the destination. The data transformation process will include the use of custom visuals to understand and work with the dataset in development mode.

Dataset3: This dataset is in a lakehouse. The data will be bulk loaded. The data transformation process will include row-based windowing functions during the loading process.

You need to identify which type of item to use for the datasets. The solution must minimize development effort and use built-in functionality, when possible.

What should you identify for each dataset? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Dataset1:	Dataflow Gen2 dataflow A notebook A T-SQL statement
Dataset2:	Dataflow Gen2 dataflow A notebook A T-SQL statement
Dataset3:	Dataflow Gen2 dataflow A KQL queryset A T-SQL statement

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

Dataset1:
 Dataflow Gen2 dataflow
 A notebook
 A T-SQL statement

Dataset2:
 Dataflow Gen2 dataflow
 A notebook
 A T-SQL statement

Dataset3:
 Dataflow Gen2 dataflow
 A KQL queryset
 A T-SQL statement

Box 1: A T-SQL statement

Dataset1: This dataset will be added to Fabric and will have a unique primary key between the source and the destination. The unique primary key will be an integer and will start from 1 and have an increment of 1.

Box 2: A Notebook

Dataset2: This dataset contains semi-structured data that uses bulk data transfer. The dataset must be handled in one process between the source and the destination. The data transformation process will include the use of custom visuals to understand and work with the dataset in development mode.

A Microsoft Fabric notebook is a primary code item for developing Apache Spark jobs and machine learning experiments. It's a web-based interactive surface used by data scientists and data engineers to write code benefiting from rich visualizations and Markdown text.

Fabric notebooks provide code snippets that help you easily write commonly used code patterns, like:

Reading data as an Apache Spark DataFrame

Drawing charts with Matplotlib

Box 3: Dataflow Gen2 dataflow

Dataset3: This dataset is in a lakehouse. The data will be bulk loaded. The data transformation process will include row-based windowing functions during the loading process.

Data Factory, Window functions in mapping data flow

QUESTION: 90

HOTSPOT (Drag and Drop is not supported)

You have a Fabric workspace that contains a lakehouse named Lakehouse1. Lakehouse1 contains a table named Status_Target that has the following columns:

Key

Status

LastModified

The data source contains a table named Status_Source that has the same columns as Status_Target.

Status_Source is used to populate Status_Target.

In a notebook name Notebook1, you load Status_Source to a DataFrame named sourceDF and

Status_Target to a DataFrame named targetDF.

You need to implement an incremental loading pattern by using Notebook1. The solution must meet the

following requirements:

For all the matching records that have the same value of key, update the value of LastModified in Status_Target to the value of LastModified in Status_Source. Insert all the records that exist in Status_Source that do NOT exist in Status_Target.

Set the value of Status in Status_Target to inactive for all the records that were last modified more than seven days ago and that do NOT exist in Status_Source.

How should you complete the statement? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

...

(targetDF

.merge(sourceDF, "sourceDF.Key" = "targetDF.Key")

☐	.whenMatchedInsert (
☐	.whenMatchedUpdate (
☐	.whenNotMatchedBySourceInsert (
☐	.whenNotMatchedBySourceUpdate (
☐	.whenNotMatchedInsert (
☐	.whenNotMatchedUpdate (

set = {"targetDF.LastModified": "sourceDF.LastModified"}

)

☐	.whenMatchedInsert (
☐	.whenMatchedUpdate (
☐	.whenNotMatchedBySourceInsert (
☐	.whenNotMatchedBySourceUpdate (
☐	.whenNotMatchedInsert (
☐	.whenNotMatchedUpdate (

values = {

"targetDF.Key": "sourceDF.Key",

"targetDF.LastModified": "sourceDF.LastModified",

"targetDF.Status": "sourceDF.Status"

}

)

☐	.whenMatchedInsert (
☐	.whenMatchedUpdate (
☐	.whenNotMatchedBySourceInsert (
☐	.whenNotMatchedBySourceUpdate (
☐	.whenNotMatchedInsert (
☐	.whenNotMatchedUpdate (

condition="targetDF.LastModified > (current_date() - INTERVAL '7' DAY)",

set = {"targetDF.Status": "'inactive'"}

)

.execute()

)

...

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

...

(targetDF

.merge(sourceDF, "sourceDF.Key" = "targetDF.Key")

.whenMatchedInsert (
.whenMatchedUpdate (
.whenNotMatchedBySourceInsert (
.whenNotMatchedBySourceUpdate (
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.whenNotMatchedUpdate (

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.whenNotMatchedBySourceUpdate (
.whenNotMatchedInsert (
.whenNotMatchedUpdate (

values = {

"targetDF.Key": "sourceDF.Key",

"targetDF.LastModified": "sourceDF.LastModified",

"targetDF.Status": "sourceDF.Status"

}

)

.whenMatchedInsert (
.whenMatchedUpdate (
.whenNotMatchedBySourceInsert (
.whenNotMatchedBySourceUpdate (
.whenNotMatchedInsert (
.whenNotMatchedUpdate (

condition="targetDF.LastModified > (current_date() - INTERVAL '7' DAY)",

set = {"targetDF.Status": "'inactive'"}

)

.execute()

)

...

Explanation:

MERGE is the most powerful operation you can do with Delta Lake. With merge, you can apply all three standard data manipulation language operations (INSERT, UPDATE, and DELETE) in a single transaction. You can also add multiple conditions to each of these operations for more complex scenarios and datasets.

Box 1: . whenMatchedUpdate(

For all the matching records that have the same value of key, update the value of LastModified in Status_Target to the value of LastModified in Status_Source.

Note: Operation semantics

Here is a detailed description of the merge programmatic operation.

There can be any number of whenMatched and whenNotMatched clauses.

whenMatched clauses are executed when a source row matches a target table row based on the match condition. These clauses have the following semantics.

whenMatched clauses can have at most one update and one delete action. The update action in merge only updates the specified columns (similar to the update operation) of the matched target row. The

delete action deletes the matched row.

Each whenMatched clause can have an optional condition. If this clause condition exists, the update or delete action is executed for any matching source-target row pair only when the clause condition is true. Etc.

Box 2: .whenNotMatchedInsert(

Insert all the records that exist in Status_Source that do NOT exist in Status_Target.

Note: whenNotMatched clauses are executed when a source row does not match any target row based on the match condition. These clauses have the following semantics.

whenNotMatched clauses can have only the insert action. The new row is generated based on the specified column and corresponding expressions. You do not need to specify all the columns in the target table. For unspecified target columns, NULL is inserted.

Box 3: .whenNotMatchedBySourceUpdate(

Set the value of Status in Status_Target to inactive for all the records that were last modified more than seven days ago and that do NOT exist in Status_Source.

Note: whenNotMatchedBySource clauses are executed when a target row does not match any source row based on the merge condition. These clauses have the following semantics.

whenNotMatchedBySource clauses can specify delete and update actions.

QUESTION: 91

DRAG DROP (Drag and Drop is not supported)

You are building a data loading pattern by using a Fabric data pipeline. The source is an Azure SQL database that contains 25 tables. The destination is a lakehouse.

In a warehouse, you create a control table named Control.Object as shown in the exhibit. (Click the Exhibit tab.)

	schema_name	table_name
1	Warehouse	ColdRoomTemperatures
2	Warehouse	Colors
3	Warehouse	PackageTypes
4	Warehouse	StockGroups
5	Warehouse	StockItems
6	dbo	BuildVersion
7	dbo	ErrorLog
8	Application	SystemParameters
9	Purchasing	PurchaseOrderLines
10	Purchasing	PurchaseOrders

You need to build a data pipeline that will support the dynamic ingestion of the tables listed in the control table by using a single execution.

Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.

Select and Place:

Actions

- Add a ForEach activity to iterate over the list of tables and copy the source data to the lakehouse Delta tables.
- Add a Get metadata activity to query Control.Object and generate a list of schemas and tables to copy.
- Add an Until activity to iterate over the list of tables and copy the source data to the lakehouse Delta tables.
- Add a Lookup activity to query Control.Object and generate a list of the schemas and tables to copy.
- Add a Copy data activity as an inner activity to the iterator activity.

Answer Area

1.
2.
3.

A. See Explanation section for answer.

Answer(s): A**Explanation:**

Actions

- Add a ForEach activity to iterate over the list of tables and copy the source data to the lakehouse Delta tables.
- Add a Get metadata activity to query Control.Object and generate a list of schemas and tables to copy.
- Add an Until activity to iterate over the list of tables and copy the source data to the lakehouse Delta tables.
- Add a Lookup activity to query Control.Object and generate a list of the schemas and tables to copy.
- Add a Copy data activity as an inner activity to the iterator activity.

Answer Area

1. Add a Lookup activity to query Control.Object and generate a list of the schemas and tables to copy.
2. Add a ForEach activity to iterate over the list of tables and copy the source data to the lakehouse Delta tables.
3. Add a Copy data activity as an inner activity to the iterator activity.

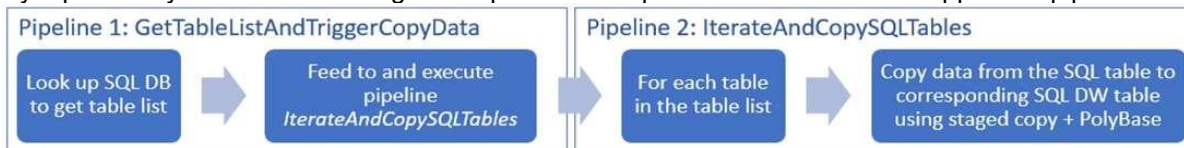
Explanation:

Step 1: Add a Lookup activity...

Copy multiple tables in bulk by using Azure Data Factory in the Azure portal

End-to-end workflow

In this scenario, you have a number of tables in Azure SQL Database that you want to copy to Azure Synapse Analytics. Here is the logical sequence of steps in the workflow that happens in pipelines:



* The first pipeline looks up the list of tables that needs to be copied over to the sink data stores. [Step 1] Alternatively you can maintain a metadata table that lists all the tables to be copied to the sink data store. Then, the pipeline triggers another pipeline, which iterates over each table in the database [Step 2] and performs the data copy operation [Step 3].

* The second pipeline performs the actual copy. It takes the list of tables as a parameter. For each table in the list, copy the specific table in Azure SQL Database to the corresponding table in Azure Synapse Analytics using staged copy via Blob storage and PolyBase for best performance. In this example, the first pipeline passes the list of tables as a value for the parameter.

Step 2: Add a ForEach activity..

The ForEach activity iterates through a list of tables, and passes one by one to the Copy activity.

Example [replace LookupTableList with Lookup Schema and Table]:

In the Parameters section, click the input box under VALUE -> select the Add dynamic content below -> enter @activity('LookupTableList').output.value as table name value -> select Finish. You're setting the result list from the Lookup activity as an input to the second pipeline. The result list contains the list of tables whose data needs to be copied to the destination.

General Settings User properties

Invoked pipeline * IterateAndCopySQLTables Open + New

Wait on completion ☐

Parameters

NAME	TYPE	VALUE	DEFAULT VALUE
tableList	array	@activity('LookupTableList').output.value	

Step 3: Add a Copy data activity..

QUESTION: 92

You are implementing a medallion architecture in a Fabric lakehouse.

You plan to create a dimension table that will contain the following columns:

ID

CustomerCode

CustomerName

CustomerAddress

CustomerLocation

ValidFrom

ValidTo

You need to ensure that the table supports the analysis of historical sales data by customer location at the time of each sale.

Which type of slowly changing dimension (SCD) should you use?

- A. Type 2
- B. Type 0
- C. Type 1
- D. Type 3

Answer(s): A

Explanation:

Slowly changing dimension type 2

With slowly changing dimension type 2, historical data is maintained by adding a new row when a dimension changes and properly denoting this new row as current while denoting the newly historical record accordingly.

Using this technique to implement slowly changing dimension type 1 not only preserves historic data, but it also offers information about when data changes. Maintaining historical data enables data analysts and data scientists to explore operational changes, perform A/B testing, and empower informed decision-making.

Incorrect:

Not C:

Slowly changing dimension type 1

Using slowly changing dimension type 1, if a record in a dimension table changes, the existing record is updated or overwritten. Otherwise, the new record is inserted into the dimension table. This means records in the dimension table always reflect the current state, and no historical data is maintained. This design approach is common for columns that store supplementary values, like the email address or phone number of a customer.

When a customer's email address or phone number changes, the dimension table updates the customer row with the new values. It's as if the customer always had this contact information.

Not D:

The Slowly Changing Dimension transformation does not support Type 3 changes, which require changes to the dimension table. By identifying columns with the fixed attribute update type, you can

capture the data values that are candidates for Type 3 changes.

QUESTION: 93

You have a Fabric workspace that contains an eventstream named EventStream1. EventStream1 outputs events to a table named Table1 in a lakehouse. The streaming data is sourced from motorway sensors and represents the speed of cars.

You need to add a transformation to EventStream1 to average the car speeds. The speeds must be grouped by non-overlapping and contiguous time intervals of one minute. Each event must belong to exactly one window.

Which windowing function should you use?

- A. sliding
- B. hopping
- C. tumbling
- D. session

Answer(s): C

Explanation:

Tumbling window

Use Tumbling window functions to segment a data stream into distinct time segments, and perform a function against them.

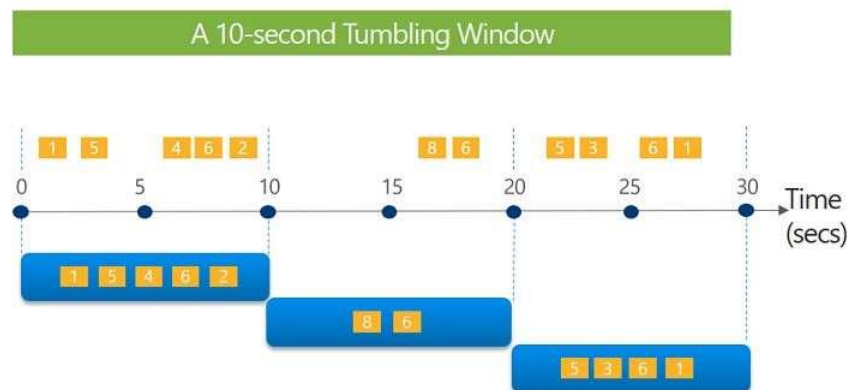
The key differentiators of a tumbling window are:

They don't repeat.

They don't overlap.

An event can't belong to more than one tumbling window.

Tell me the count of Tweets per time zone every 10 seconds



```
SELECT TimeZone, COUNT(*) AS Count
FROM TwitterStream TIMESTAMP BY CreatedAt
GROUP BY TimeZone, TumblingWindow(second,10)
```

QUESTION: 94

HOTSPOT (Drag and Drop is not supported)

You have a table in a Fabric lakehouse that contains the following data.

SalesOrderNumber	OrderDate	CustomerName	Email
SO49172	2021-01-01	Brian Howard	brian23@adventure-works.com
SO49173	2021-01-01	Linda Alvarez	linda19@adventure-works.com
SO49174	2021-01-01	Gina Hernandez	gina4@adventure-works.com
SO49178	2021-01-01	Beth Ruiz	beth4@adventure-works.com
SO49179	2021-01-01	Evan Ward	evan13@adventure-works.com

You have a notebook that contains the following code segment.

```
01 df = df.withColumn("CustomerName", when((col("CustomerName").isNull() | (col("CustomerName")=="")),lit("Unknown")).otherwise(col("CustomerName")))
02 df = df.withColumn("Username",split(col("Email"), "@").getItem(1))
03 df = df.dropDuplicates(["OrderDate"]).select(col("OrderDate"), year("OrderDate").alias("Year"), ("CustomerName"), ("Username"))
04 display(df.head(10))
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements

Line 01 will replace all the null and empty values in the CustomerName column with the Unknown value.

Yes

☐

No

☐

Line 02 will extract the value before the @ character and generate a new column named Username.

☐
☐

Line 03 will extract the year value from the OrderDate column and keep only the first occurrence for each year.

☐
☐

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

Statements

Line 01 will replace all the null and empty values in the CustomerName column with the Unknown value.

Yes

☒

No

☐

Line 02 will extract the value before the @ character and generate a new column named Username.

☐
☒

Line 03 will extract the year value from the OrderDate column and keep only the first occurrence for each year.

☒
☐

Box 1: Yes

Yes - Line 01 will replace all the null and empty values in the CustomerName column with the Unknown value.

Line 01 contains an isnull(), the logical operator or (|), and check for an empty value (="").

Box 2: No

No - Line 02 will extract the value before the @ character and generate a new column named Username.

Line 02: df = df.withColumn("Username",split(col("Email"),"@").getItem(1))

The extracted value is after the @ character, not before. getItem(0) would have extracted the value before the @ character.

Note: pyspark.sql.functions provide a function split() which is used to split DataFrame string Column into multiple columns.

Box 3: Yes

Yes - Line 03 will extract the year value from the OrderDate column and keep only the first occurrence for each year.

Note: pyspark.sql.DataFrame.dropDuplicates

DataFrame.dropDuplicates(subset: Optional[List[str]] = None) pyspark.sql.dataframe.DataFrame[source]

Return a new DataFrame with duplicate rows removed, optionally only considering certain columns.

QUESTION: 95

You have a Fabric workspace that contains a warehouse named Warehouse1.

You have an on-premises Microsoft SQL Server database named Database1 that is accessed by using an on-premises data gateway.

You need to copy data from Database1 to Warehouse1.

Which item should you use?

- A. a data pipeline
- B. an Apache Spark job definition
- C. a streaming dataflow
- D. a notebook

Answer(s): A

Explanation:

How to access on-premises data sources in Data Factory for Microsoft Fabric You can import On Premise SQL Server 2022 Data into Microsoft Fabric with Data Pipelines,

Note: Using on-premises data in a pipeline

1. Go to your workspace and create a data pipeline.
2. From the Home tab of the pipeline editor, select Copy data and then Use copy assistant. Add a new source to the activity in the assistant's Choose data source page, then select the connection established in the previous step.
3. Select a destination for your data from the on-premises data source.
4. Run the pipeline.

Now you've created and ran a pipeline to load data from an on-premises data source into a cloud destination.

QUESTION: 96

You have a Fabric warehouse named DW1 that contains a Type 2 slowly changing dimension (SCD) dimension table named DimCustomer. DimCustomer contains 100 columns and 20 million rows. The columns are of various data types, including int, varchar, date, and varbinary.

You need to identify incoming changes to the table and update the records when there is a change. The solution must minimize resource consumption.

What should you use to identify changes to attributes?

- A. a hash function to compare the attributes in the source table.

- B. a direct attributes comparison across the attributes in the DimCustomer table.
- C. a direct attributes comparison for the attributes in the source table.
- D. a hash function to compare the attributes in the DimCustomer table.

Answer(s): D

Explanation:

Implementing a Type 2 Slowly Changing Dimension (SCD) with a hash value Implementing a Type 2 Slowly Changing Dimension (SCD) with a hash value is a common technique used to efficiently track changes in dimension data while reducing the need for excessive updates and comparisons. In this approach, you use a hash value to quickly determine if a record has changed, and if it has, you create a new version of the record with updated attributes.

QUESTION: 97

You have an Azure SQL database named DB1.

In a Fabric workspace, you deploy an eventstream named EventStreamDB1 to stream record changes from DB1 into a lakehouse.

You discover that events are NOT being propagated to EventStreamDB1.

You need to ensure that the events are propagated to EventStreamDB1.

What should you do?

- A. Create a read-only replica of DB1.
- B. Create an Azure Stream Analytics job.
- C. Enable Extended Events for DB1.
- D. Enable change data capture (CDC) for DB1.

Answer(s): D

Explanation:

Build real-time order notifications with Eventstream's CDC connector Change Data Capture (CDC) is a popular database feature for monitoring and recording all row-level changes to a table. It plays a significant role in handling transactional events, such as sales orders from an online store, and providing real-time status updates to customers. With Eventstream's CDC connector, you can now achieve this seamlessly, capturing changes in your database in real time. You can then transform the changed data in Eventstream and route it to various destinations within Fabric for alerting and analysis.

QUESTION: 98

DRAG DROP (Drag and Drop is not supported)

You have a Fabric workspace that contains an eventhouse named Eventhouse1.

In Eventhouse1, you plan to create a table named DeviceStreamData in a KQL database. The table will contain data based on the following sample.

Timestamp	DeviceId	StreamData
2024-05-18 12:45:17.16524	81416f30-60a2-4e75-9b19-2a84ea059735	[{ "index": 0, "eventId": "719afca0-be30-4559-bb5e-59feade642f6", "isActive": false, "latitude": 5.390012, "longitude": -40.100235, "tags": ["tempor"] }]
2024-05-18 12:45:21.76423	bb664e1e-02aa-4e17-8c8a-116cd4458d52	[{ "index": 0, "eventId": "782222b2-fbcb-43c0-82d6-ecd49a99dbf5", "isActive": true, "latitude": -56.153786, "longitude": 130.870907, "tags": ["adipisicing"] }]
2024-05-18 12:45:23.98642	717bfe7d-0e5d-498f-9f21-e60aaf258056	[{ "index": 0, "eventId": "d5730286-0da4-41f8-8e59-f75e209310a9", "isActive": true, "latitude": -21.39289, "longitude": 123.959442, "tags": ["ad"] }]
2024-05-18 12:45:25.39523	1a390e71-4faf-4df5-a479-2238d84001f7	[{ "index": 0, "eventId": "9572e141-8692-4d16-89e8-002f9b7e22b6", "isActive": true, "latitude": -84.926214, "longitude": -11.499007, "tags": ["ex"] }]
2024-05-18 12:45:27.43343	2f0ba7d0-6dff-4081-bf7f-d0d39d4c3260	[{ "index": 0, "eventId": "08a42b87-ce84-4bb2-99f0-4fb5c75ff63f", "isActive": true, "latitude": -49.909339, "longitude": -177.505775, "tags": ["laboris"] }]

You need to use a KQL query to develop the solution for Eventhouse1.

Which three code segments should you run in sequence? To answer, move the appropriate code segments from the list of code segments to the answer area and arrange them in the correct order. Select and Place:

Code Segments

StreamData:long)

.create table EventStreamData (

StreamData:dynamic)

.create function EventStreamData (

TimeStamp:datetime, DeviceId:string

Answer Area

1.

2.

3.

A. See Explanation section for answer.

Answer(s): A

Explanation:

Code Segments

```
StreamData:long)

.create table EventStreamData (

StreamData:dynamic )

.create function EventStreamData (

TimeStamp:datetime, DeviceId:string
```

Answer Area

```
1. .create table EventStreamData (
2. TimeStamp:datetime, DeviceId:string
3. StreamData:dynamic )
```

Explanation:

Step 1: .create table EventStreamData (

Prepare your Eventhouse environment, example

In this setup step, you create a table in an Eventhouse with the necessary columns and encoding policies to store the vector data.

1. Browse to your workspace homepage in Real-Time Intelligence.
2. Select the Eventhouse you created in the prerequisites.
3. Select the target database where you want to store the vector data. If you don't have a database, you can create one by selecting Add database.
4. Select Explore my data. Copy/paste the following KQL query to create a table with the necessary columns:

```
.create table Wiki (id:string,url:string,
['title']:string,text:string,title_vector:dynamic,content_vector:dynamic,vector_id:long)
```

Incorrect:

```
* .create function EventStreamData (
Step 2: TimeStamp:datetime, DeviceID:string
Step 3: StreamData:dynamic)
```

Incorrect:

```
* StreamData:long)
```

QUESTION: 99

DRAG DROP (Drag and Drop is not supported)

You have a KQL database that contains a table named Readings.

You need to build a KQL query to compare the MeterReading value of each row to the previous row based on the Timestamp value.

A sample of the expected output is shown in the following table.

City	Area	MeterReading	Timestamp	PrevMeterReading	PrevTimestamp
Kansas	Area1	1500	2024-07-30 10:00:00		
Kansas	Area2	1520	2024-07-30 11:00:00	1500	2024-07-30 10:00:00
Kansas	Area1	1550	2024-07-30 12:00:00	1520	2024-07-30 11:00:00
Kansas	Area2	1580	2024-07-30 13:00:00	1550	2024-07-30 12:00:00

How should you complete the query? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

Note: Each correct selection is worth one point.

Select and Place:

Values	Answer Area
evaluate	Readings
extend	filter City == "Kansas"
lookup	by Timestamp
project	PrevMeterReading = prev(MeterReading),
sort	PrevTimestamp = prev(Timestamp)
summarize	City, Area, MeterReading, Timestamp, PrevMeterReading, PrevTimeStamp
take	

A. See Explanation section for answer.

Answer(s): A

Explanation:

Values	Answer Area
evaluate	Readings
extend	filter City == "Kansas"
lookup	sort by Timestamp
project	extend PrevMeterReading = prev(MeterReading),
sort	PrevTimestamp = prev(Timestamp)
summarize	project City, Area, MeterReading, Timestamp, PrevMeterReading, PrevTimeStamp
take	

Box 1: sort

Need to sort on the Timestamp column.

Box 2: extend

Filter data based on comparison between adjacent rows

The following query returns rows that show breaks longer than a quarter of a second between calls to sensor-9.

TransformedSensorsData

| where SensorName == 'sensor-9'

| sort by Timestamp asc

| extend timeDiffInMilliseconds = datetime_diff('millisecond', Timestamp, prev(Timestamp, 1)) | where timeDiffInMilliseconds > 250

Box 3: project

Extend row with data from the previous row

In the following query, as part of the serialization done with the serialize operator, a new column previous_session_type is added with data from the previous row. Since there was no session prior to the first session, the column is empty in the first row.

ConferenceSessions

```
| where conference == 'Build 2019'
| serialize previous_session_type = prev(session_type)
| project time_and_duration, session_title, session_type, previous_session_type
```

QUESTION: 100

HOTSPOT (Drag and Drop is not supported)

You need to recommend a Fabric streaming solution that will use the sources shown in the following table.

Name	Message size	Description
Source1	10 MB	Contains semi-structured data that has a bigint column in the messages
Source2	25 MB	Contains structured data that has 19 columns
Source3	5 MB	Contains unstructured data that has images in the messages

The solution must minimize development effort.

What should you include in the recommendation for each source? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Source1:

- Apache Spark Structured Streaming
- An eventstream
- A data pipeline
- A streaming dataflow

Source2:

- Apache Spark Structured Streaming
- An eventstream
- A data pipeline
- A streaming dataflow

Source3:

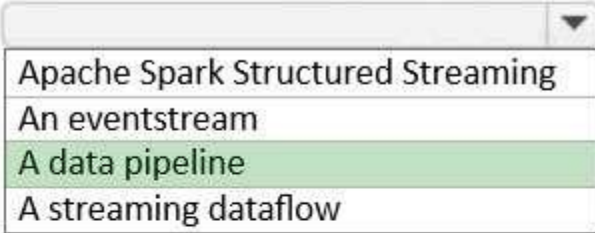
- Apache Spark Structured Streaming
- An eventstream
- A data pipeline
- A streaming dataflow

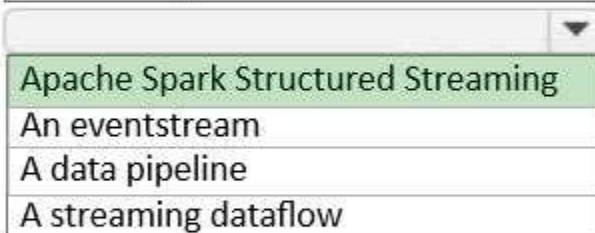
A. See Explanation section for answer.

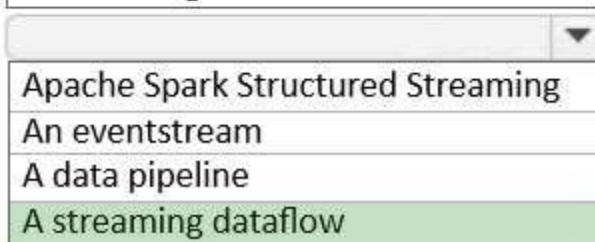
Answer(s): A

Explanation:

Answer Area

Source1: 

Source2: 

Source3: 

Box 1: A data pipeline

Source1: Message size 10 MB, contains semi-structured data has a bigint column in the messages.

A data pipeline is a sequence of data processing steps that makes the extraction, transformation, and loading (ETL) of data from different sources to a destination possible, such as a data warehouses etc.

Data Sources(Initial)

The origins from which data is collected. These can be diverse and include structured, *semi-structured*, and unstructured data sources.

Box 2: Apache Spark Structured Streaming

Source2: Message size 25 MB, contains structured data has 19 columns

Box 3: A streaming dataflow

Source3: Message size 5 MB, contains unstructured data has images in the messages.

Streaming dataflows

Streaming dataflows allow authors to connect to, ingest, mash up, model, and build reports based on streaming in near real-time data directly in the Power BI service.

Blob storage

Azure Blob storage is Microsoft's object storage solution for the cloud. Blob storage is optimized for storing massive amounts of unstructured data. Unstructured data is data that doesn't adhere to a particular data model or definition, such as text or binary data.

You can use Azure Blobs as a streaming or reference input. Streaming blobs are checked every second for updates. Unlike a streaming blob, a reference blob is only loaded at the beginning of the refresh. It's static data that isn't expected to change, and the recommended limit for static data is 50 MB or less.

Incorrect:

* An eventstream

Fabric Eventstream has the following general limitations. Before working with Eventstream, review these limitations to ensure they align with your requirements.

*-> Maximum message size: 1 MB

* Maximum retention period of event data: 90 days

QUESTION: 101

HOTSPOT (Drag and Drop is not supported)

You are building a data loading pattern for Fabric notebook workloads.

You have the following code segment.

```
def loading_pattern_sample(df_source):
    try:
        deltaTable = DeltaTable.forName(spark, target_table)
    except Exception:
        try:
            df_source.write.format('delta').mode('overwrite').saveAsTable(f"{target_table}")
        except Exception as e:
            print(f'Load for table {target_table} failed with error: {str(e)}')
            raise
        return

    try:
        change_detection_columns = [col for col in df_source.columns if col not in candidate_key]

        match_condition = ' AND '.join([f'target.{col} = source.{col}' for col in candidate_key])
        update_condition = ' OR '.join([f'target.{col} != source.{col}' for col in change_detection_columns])

        update_expr = {col: f'source.{col}' for col in df_source.columns}

        merge_operation = deltaTable.alias('target').merge(
            source=df_source.alias('source'),
            condition=match_condition
        ).whenMatchedUpdate(
            condition=update_condition,
            set=update_expr
        ).whenNotMatchedInsertAll()

        merge_operation.execute()
    except Exception as e:
        print(f'Insert operation for table {target_table} failed with error: {str(e)}')
    return
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
The target table will always be overwritten.	☐	☐
The merge operation will always run.	☐	☐
The code supports incremental loading and the initial table creation.	☐	☐

A. See Explanation section for answer.

Answer(s): A**Explanation:****Answer Area**

Statements	Yes	No
The target table will always be overwritten.	☐	☒
The merge operation will always run.	☐	☒
The code supports incremental loading and the initial table creation.	☒	☐

Box 1: No

The target table will always be overwritten

The code first checks if the target Delta table exists using `DeltaTable.forName`. If it does not exist, the table is created, and data is written using `df_source.write.format('delta').mode('overwrite')`. However, if the table exists, a merge operation is executed instead of overwriting the table. Therefore, the target table is not always overwritten.

Box 2: No

The merge operation will always run

The merge operation only runs if the target Delta table already exists. If the table does not exist, the data is written directly without performing a merge. Therefore, the merge operation does not always run.

Box 3: Yes

The code supports incremental loading and the initial table creation.

The loading pattern supports full loads by overwriting the table when it does not exist and incremental loads through the merge operation. The merge operation detects changes (via `update_condition`) and inserts or updates records accordingly. This supports both full and incremental loading.

QUESTION: 102

You have an Azure event hub. Each event contains the following fields:

BikepointID

Street

Neighbourhood

Latitude

Longitude

No_Bikes

No_Empty_Docks

You need to ingest the events. The solution must only retain events that have a Neighbourhood value of Chelsea, and then store the retained events in a Fabric lakehouse.

Data retention in case of failure is required to be two days.

What should you use?

- A. an eventstream
- B. Apache Spark Structured Streaming
- C. a streaming dataset
- D. a KQL queryset

Answer(s): B**Explanation:**

An eventstream is the best solution for ingesting data from Azure Event Hub into Fabric, while applying filtering logic such as retaining only the events that have a Neighbourhood value of "Chelsea."

Eventstreams in Microsoft Fabric are designed for handling real-time data streams and can apply transformation logic directly on incoming events. In this case, the eventstream can filter events based on the Neighbourhood field before storing the retained events in a Fabric lakehouse.

Eventstreams are well-suited for stream processing, such as this case where you need to filter out only specific data (events with a Neighbourhood of "Chelsea") before storing it in the lakehouse.

Retention setting

For the retention setting, you can specify the duration for which the incoming data needs to be retained.

The default retention period is one day. Events are automatically removed when the retention period expires. If you set the retention period to one day (24 hours), the event becomes unavailable exactly 24 hours after it's accepted. You can't explicitly delete events. The maximum value for this setting is 90 days.

Reference:

<https://learn.microsoft.com/en-us/fabric/real-time-intelligence/event-streams/configure-settings>

QUESTION: 103

You have a Fabric workspace that contains a lakehouse named Lakehouse1.

In an external data source, you have data files that are 500 GB each. A new file is added every day.

You need to ingest the data into Lakehouse1 without applying any transformations. The solution must meet the following requirements:

Trigger the process when a new file is added.

Provide the highest throughput.

Which type of item should you use to ingest the data?

- A. KQL queryset
- B. Streaming dataset
- C. Notebook
- D. Dataflow Gen2

Answer(s): C

Explanation:

Correct:

* Data pipeline [Best]

To efficiently ingest large data files (500 GB each) into Lakehouse1 with high throughput and trigger the process when a new file is added, a Data pipeline is the most suitable solution. Data pipelines in Fabric are ideal for orchestrating data movement and can be configured to automatically trigger based on file arrivals or other events. This solution meets both requirements: ingesting the data without transformations (since you just need to copy the data) and triggering the process when new files are added.

* Notebook [Second best]

You can create the Data Pipeline that will load the raw data into the Lakehouse Files area and trigger the notebook to load the raw files into the Lakehouse Tables area as Delta tables.

Incorrect:

* Dataflow Gen2

* Environment

* KQL queryset

* Streaming dataset

Note: Different ways to load data into a lakehouse

In Microsoft Fabric, there are a few ways you can get data into a lakehouse:

* File upload from local computer

* Run a copy tool in pipelines [Data pipeline - correct]

Copy tool in pipelines

The Copy tool is a highly scalable Data Integration solution that allows you to connect to different data sources and load the data either in original format or convert it to a Delta table. Copy tool is a part of pipelines activities that you can modify in multiple ways, such as scheduling or triggering based on an event.

* Set up a dataflow

* Apache Spark libraries in notebook code [Correct in conjunction with a data pipeline] You can use available Spark libraries to connect to a data source directly, load data to a data frame, and then save it in a lakehouse. This method is the most open way to load data in the lakehouse that user code is fully managing.

* Stream real-time events with Eventstream

With Eventstream, you can get, process, and route high volumes real-time events from a wide variety of sources.

* Get data from Eventhouse

When you enable OneLake availability on data in an Eventhouse, a Delta table is created in OneLake. This Delta table can be accessed by a lakehouse using a shortcut.

Note 2: Considerations when choosing approach to load data

Use case	Recommendation
Small file upload from local machine	Use Local file upload
Small data or specific connector	Use Dataflows
Large data source	Use Copy tool in pipelines
Complex data transformations	Use Notebook code
Streaming data	Use Eventstream to stream data into Eventhouse; enable OneLake availability and create a shortcut from Lakehouse
Time-series data	Get data from Eventhouse

Reference:

<https://learn.microsoft.com/en-us/fabric/data-engineering/load-data-lakehouse>

Exam Topic: Monitor and optimize an analytics solution (Testlet 1)

Case Study:

This is a case study. Case studies are not timed separately. You can use as much exam time as you would like to complete each case. However, there may be additional case studies and sections on this exam.

You must manage your time to ensure that you are able to complete all questions included on this exam in the time provided.

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questions in this case study.

At the end of this case study, a review screen will appear. This screen allows you to review your answers and to make changes before you move to the next section of the exam. After you begin a new section, you cannot return to this section.

To start the case study

To display the first question in this case study, click the Next button. Use the buttons in the left pane to explore the content of the case study before you answer the questions. Clicking these buttons displays information such as business requirements, existing environment, and problem statements. If the case study has an All Information tab, note that the information displayed is identical to the information displayed on the subsequent tabs.

When you are ready to answer a question, click the Question button to return to the question.

Overview:

Litware, Inc. is a publishing company that has an online bookstore and several retail bookstores worldwide.

Litware also manages an online advertising business for the authors it represents.

Existing Environment. Fabric Environment

Litware has a Fabric workspace named Workspace1. High concurrency is enabled for Workspace1. The company has a data engineering team that uses Python for data processing.

Existing Environment. Data Processing

The retail bookstores send sales data at the end of each business day, while the online bookstore constantly provides logs and sales data to a central enterprise resource planning (ERP) system. Litware implements a medallion architecture by using the following three layers: bronze, silver, and gold. The sales data is ingested from the ERP system as Parquet files that land in the Files folder in a lakehouse.

Notebooks are used to transform the files in a Delta table for the bronze and silver layers. The gold layer is in a warehouse that has V-Order disabled.

Litware has image files of book covers in Azure Blob Storage. The files are loaded into the Files folder.

Existing Environment. Sales Data

Month-end sales data is processed on the first calendar day of each month. Data that is older than one month never changes.

In the source system, the sales data refreshes every six hours starting at midnight each day.

The sales data is captured in a Dataflow Gen2 dataflow.

When the dataflow runs, new and historical data is captured. The dataflow captures the following fields of the source:

Sales Date

Author

Price

Units

SKU

A table named AuthorSales stores the sales data that relates to each author. The table contains a column named AuthorEmail. Authors authenticate to a guest Fabric tenant by using their email address.

Existing Environment. Security Groups

Litware has the following security groups:

Sales

Fabric Admins

Streaming Admins

Existing Environment. Performance Issues

Business users perform ad-hoc queries against the warehouse. The business users indicate that reports against the warehouse sometimes run for two hours and fail to load as expected. Upon further investigation, the data engineering team receives the following error message when the reports fail to load: "The SQL query failed while running."

The data engineering team wants to debug the issue and find queries that cause more than one failure. When the authors have new book releases, there is often an increase in sales activity. This increase slows the data ingestion process.

The company's sales team reports that during the last month, the sales data has NOT been up-to-date when they arrive at work in the morning.

Requirements. Planned Changes

Litware recently signed a contract to receive book reviews. The provider of the reviews exposes the data in Amazon Simple Storage Service (Amazon S3) buckets.

Litware plans to manage Search Engine Optimization (SEO) for the authors. The SEO data will be streamed from a REST API.

Requirements. Version Control

Litware plans to implement a version control solution in Fabric that will use GitHub integration and follow the principle of least privilege.

Requirements. Governance Requirements

To control data platform costs, the data platform must use only Fabric services and items. Additional Azure resources must NOT be provisioned.

Requirements. Data Requirements

Litware identifies the following data requirements:

Process the SEO data in near-real-time (NRT).

Make the book reviews available in the lakehouse without making a copy of the data.

When a new book cover image arrives in the Files folder, process the image as soon as possible.

QUESTION: 104

What should you do to optimize the query experience for the business users?

- A. Enable V-Order.
- B. Create and update statistics.
- C. Run the VACUUM command.
- D. Introduce primary keys.

Answer(s): B

Explanation:

Microsoft Fabric, Troubleshoot the Warehouse

Query performance seems to degrade over time

Many factors can affect a query's performance, such as changes in table size, data skew, workload concurrency, available resources, network, etc. Just because a query runs slower doesn't necessarily mean there's a query performance problem. Take following steps to investigate the target query:

1. Identify the differences in all performance-affecting factors among good and bad performance runs.

*-> 2. Verify proper column statistics have been created on all tables.

*-> 3. Ensure all table statistics are updated after large DML transactions.

4. Check for data skew in base tables.

5. Pause and resume the service. Then, rerun the query when there's no other active queries running.

You can monitor the warehouse workload using DMV.

Scenario:

Existing Environment. Performance Issues

Business users perform ad-hoc queries against the warehouse. The business users indicate that reports against the warehouse sometimes run for two hours and fail to load as expected. Upon further investigation, the data engineering team receives the following error message when the reports fail to load: "The SQL query failed while running."

QUESTION: 105

HOTSPOT (Drag and Drop is not supported)

You need to troubleshoot the ad-hoc query issue.

How should you complete the statement? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

```
SELECT last_run_start_time, last_run_command
```

FROM

queryinsights.exec_requests_history
queryinsights.exec_sessions_history
queryinsights.frequently_run_queries
queryinsights.long_running_queries

```
WHERE last_run_total_elapsed_time_ms > 7200000
```

AND

max_run_total_elapsed_time_ms > 7200000
median_total_elapsed_time_ms > 7200000
number_of_canceled_runs > 1
number_of_failed_runs > 1
number_of_runs > 1

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

SELECT last_run_start_time, last_run_command

FROM

queryinsights.exec_requests_history
queryinsights.exec_sessions_history
queryinsights.frequently_run_queries
queryinsights.long_running_queries

WHERE last_run_total_elapsed_time_ms > 7200000

AND

max_run_total_elapsed_time_ms > 7200000
median_total_elapsed_time_ms > 7200000
number_of_canceled_runs > 1
number_of_failed_runs > 1
number_of_runs > 1

Explanation:

SELECT last_run_start_time, last_run_command: These fields will help identify the execution details of the long-running queries.

FROM queryinsights.long_running_queries: The correct solution is to check the long-running queries using the queryinsights.long_running_queries view, which provides insights into queries that take longer than expected to execute.

WHERE last_run_total_elapsed_time_ms > 7200000: This condition filters queries that took more than 2 hours to complete (7200000 milliseconds), which is relevant to the issue described. AND

number_of_failed_runs > 1: This condition is key for identifying queries that have failed more than once, helping to isolate the problematic queries that cause failures and need attention.

Exam Topic: Monitor and optimize an analytics solution (Testlet 2)

Case Study:

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When you are ready to answer a question, click the Question button to return to the question.

Overview:

- Company Overview

Contoso, Ltd. is an online retail company that wants to modernize its analytics platform by moving to Fabric. The company plans to begin using Fabric for marketing analytics.

Overview:

- IT Structure

The company's IT department has a team of data analysts and a team of data engineers that use analytics systems.

The data engineers perform the ingestion, transformation, and loading of data. They prefer to use Python or SQL to transform the data.

The data analysts query data and create semantic models and reports. They are qualified to write queries in Power Query and T-SQL.

Existing Environment. Fabric

Contoso has an F64 capacity named Cap1. All Fabric users are allowed to create items.

Contoso has two workspaces named WorkspaceA and WorkspaceB that currently use Pro license mode.

Existing Environment. Source Systems

Contoso has a point of sale (POS) system named POS1 that uses an instance of SQL Server on Azure Virtual Machines in the same Microsoft Entra tenant as Fabric. The host virtual machine is on a private virtual network that has public access blocked. POS1 contains all the sales transactions that were processed on the company's website.

The company has a software as a service (SaaS) online marketing app named MAR1. MAR1 has seven entities. The entities contain data that relates to email open rates and interaction rates, as well as website interactions. The data can be exported from MAR1 by calling REST APIs. Each entity has a different endpoint.

Contoso has been using MAR1 for one year. Data from prior years is stored in Parquet files in an Amazon Simple Storage Service (Amazon S3) bucket. There are 12 files that range in size from 300 MB to 900 MB and relate to email interactions.

Existing Environment. Product Data

POS1 contains a product list and related data. The data comes from the following three tables:

Products

ProductCategories

ProductSubcategories

In the data, products are related to product subcategories, and subcategories are related to product categories.

Existing Environment. Azure

Contoso has a Microsoft Entra tenant that has the following mail-enabled security groups:

DataAnalysts: Contains the data analysts

DataEngineers: Contains the data engineers

Contoso has an Azure subscription.

The company has an existing Azure DevOps organization and creates a new project for repositories that relate to Fabric.

Existing Environment. User Problems

The VP of marketing at Contoso requires analysis on the effectiveness of different types of email content. It typically takes a week to manually compile and analyze the data. Contoso wants to reduce the time to less than one day by using Fabric.

The data engineering team has successfully exported data from MAR1. The team experiences transient connectivity errors, which causes the data exports to fail.

Requirements. Planned Changes

Contoso plans to create the following two lakehouses:

Lakehouse1: Will store both raw and cleansed data from the sources

Lakehouse2: Will serve data in a dimensional model to users for analytical queries

Additional items will be added to facilitate data ingestion and transformation.

Contoso plans to use Azure Repos for source control in Fabric.

Requirements. Technical Requirements

The new lakehouses must follow a medallion architecture by using the following three layers: bronze, silver, and gold. There will be extensive data cleansing required to populate the MAR1 data in the silver layer, including deduplication, the handling of missing values, and the standardizing of capitalization. Each layer must be fully populated before moving on to the next layer. If any step in populating the lakehouses fails, an email must be sent to the data engineers.

Data imports must run simultaneously, when possible.

The use of email data from the Amazon S3 bucket must meet the following requirements:

Minimize egress costs associated with cross-cloud data access.

Prevent saving a copy of the raw data in the lakehouses.

Items that relate to data ingestion must meet the following requirements:

The items must be source controlled alongside other workspace items.

Ingested data must land in the bronze layer of Lakehouse1 in the Delta format.

No changes other than changes to the file formats must be implemented before the data lands in the bronze layer.

Development effort must be minimized and a built-in connection must be used to import the source data.

In the event of a connectivity error, the ingestion processes must attempt the connection again.

Lakehouses, data pipelines, and notebooks must be stored in WorkspaceA. Semantic models, reports, and dataflows must be stored in WorkspaceB.

Once a week, old files that are no longer referenced by a Delta table log must be removed.

Requirements. Data Transformation

In the POS1 product data, ProductID values are unique. The product dimension in the gold layer must include only active products from product list. Active products are identified by an IsActive value of 1.

Some product categories and subcategories are NOT assigned to any product. They are NOT analytically relevant and must be omitted from the product dimension in the gold layer.

Requirements. Data Security

Security in Fabric must meet the following requirements:

The data engineers must have read and write access to all the lakehouses, including the underlying files.

The data analysts must only have read access to the Delta tables in the gold layer.

The data analysts must NOT have access to the data in the bronze and silver layers.

The data engineers must be able to commit changes to source control in WorkspaceA.

QUESTION: 106

You need to schedule the population of the medallion layers to meet the technical requirements.

What should you do?

- A. Schedule a data pipeline that calls other data pipelines.
- B. Schedule a notebook.
- C. Schedule an Apache Spark job.

D. Schedule multiple data pipelines.

Answer(s): A

Explanation:

The technical requirements specify that:

Medallion layers must be fully populated sequentially (bronze silver gold). Each layer must be populated before the next.

If any step fails, the process must notify the data engineers.

Data imports should run simultaneously when possible.

Why Use a Data Pipeline That Calls Other Data Pipelines?

A data pipeline provides a modular and reusable approach to orchestrating the sequential population of medallion layers.

By calling other pipelines, each pipeline can focus on populating a specific layer (bronze, silver, or gold), simplifying development and maintenance.

A parent pipeline can handle:

- Sequential execution of child pipelines.
- Error handling to send email notifications upon failures.
- Parallel execution of tasks where possible (e.g., simultaneous imports into the bronze layer).

QUESTION: 107

HOTSPOT (Drag and Drop is not supported)

You need to ensure that the data engineers are notified if any step in populating the lakehouses fails. The solution must meet the technical requirements and minimize development effort.

What should you use? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

To identify the failure:

	▼
A Fail activity	
An If condition activity	
An On failure dependency condition	
An On completion dependency condition	

To send the notification:

	▼
A Teams activity	
An Invoke pipeline activity	
An Office365Outlook activity	

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

To identify the failure:

	▼
A Fail activity	
An If condition activity	
An On failure dependency condition	
An On completion dependency condition	

To send the notification:

	▼
A Teams activity	
An Invoke pipeline activity	
An Office365Outlook activity	

Explanation:**Scenario:****Requirements.** Technical Requirements

The new lakehouses must follow a medallion architecture by using the following three layers: bronze, silver, and gold. There will be extensive data cleansing required to populate the MAR1 data in the silver layer, including deduplication, the handling of missing values, and the standardizing of capitalization.

*-> Each layer must be fully populated before moving on to the next layer. If any step in populating the lakehouses fails, an *email* must be sent to the data engineers.

Box 1: An On failure dependency condition

To identify the failure

In Azure Data Factory pipelines, an "on failure" dependency condition specifies that a subsequent activity should run only if the preceding activity fails. This allows for error handling and fallback mechanisms within the pipeline. If an activity is linked to another activity via an "on failure" dependency, and the preceding activity fails, the linked activity will be executed. If the preceding activity succeeds, the linked activity will be skipped

Incorrect:

* An On completion dependency condition

In Azure Data Factory, the "On Completion" dependency condition ensures that the next activity in a pipeline will start regardless of whether the previous activity succeeded or failed, as long as the previous activity has finished its execution. This means the next activity will be triggered when the previous activity completes, regardless of the outcome.

Box 2: An Office365Outlook activity

To send the notification

Note: Automate and send notifications with Data Factory

For example, you can use an Office 365 Outlook activity to send the output of a Copy activity by email.

Exam Topic: Monitor and optimize an analytics solution (Question Set 3)**QUESTION: 108**

DRAG DROP (Drag and Drop is not supported)

You have a Fabric eventhouse that contains a KQL database. The database contains a table named TaxiData.

The following is a sample of the data in TaxiData.

VendorID	tpep_pickup_datetime	tpep_dropoff_datetime	passenger_count	trip_distance	PULocationID	DOLocationID	payment_type	total_amount
2	2022-06-06T11:08:32Z	2022-06-06T11:22:17Z	1	0.17	231	50	2	7.12
2	2022-06-06T11:12:05Z	2022-06-06T11:20:43Z	1	1.02	161	163	1	10.56
1	2022-06-06T11:15:00Z	2022-06-06T11:25:32Z	1	1.07	142	230	2	17.12
2	2022-06-06T11:29:54Z	2022-06-06T11:49:34Z	2	2.07	162	236	2	12.01
1	2022-06-06T11:50:50Z	2022-06-06T12:07:24Z	2	2.65	140	142	1	7.89

You need to build two KQL queries. The solution must meet the following requirements:

One of the queries must partition RunningTotalAmount by VendorID.

The other query must create a column named FirstPickupDateTime that shows the first value of each hour from tpep_pickup_datetime partitioned by payment_type.

How should you complete each query? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

Note: Each correct selection is worth one point.

Select and Place:

Values	Answer Area
Row_cumsum	Statement1:
Row_rank_dense	TaxiData
Row_rank_min	sort by VendorID asc
Row_window_session	extend RunningTotalAmount = [] (total_amount, VendorID != prev(VendorID))
	Statement2:
	TaxiData
	sort by tpep_pickup_datetime asc, payment_type asc
	extend FirstPickupDateTime = [] (tpep_pickup_datetime, 1h, 0m, payment_type != prev(payment_type))

A. See Explanation section for answer.

Answer(s): A

Explanation:

Values	Answer Area
Row_cumsum	Statement1:
Row_rank_dense	TaxiData
Row_rank_min	sort by VendorID asc
Row_window_session	extend RunningTotalAmount = [Row_cumsum] (total_amount, VendorID != prev(VendorID))
	Statement2:
	TaxiData
	sort by tpep_pickup_datetime asc, payment_type asc
	extend FirstPickupDateTime = [Row_window_session] (tpep_pickup_datetime, 1h, 0m, payment_type != prev(payment_type))

Explanation:

Partition the RunningTotalAmount by VendorID. - Row_cumsum

The Row_cumsum function computes the cumulative sum of a column while optionally restarting the accumulation based on a condition. In this case, it calculates the cumulative sum of total_amount for each VendorID, restarting when the VendorID changes (VendorID != prev(VendorID)).

TaxiData

| sort by VendorID asc

| extend RunningTotalAmount = Row_cumsum(total_amount, VendorID != prev(VendorID))

Create a column FirstPickupDateTime that shows the first value of each hour from tpep_pickup_datetime, partitioned by payment_type - Row_window_session

TaxiData

| sort by tpep_pickup_datetime asc, payment_type asc

| extend FirstPickupDateTime = Row_window_session(tpep_pickup_datetime, 1h, 0m, payment_type != prev(payment_type))

QUESTION: 109

HOTSPOT (Drag and Drop is not supported)

You are processing streaming data from an external data provider.

You have the following code segment.

```
datatable (Location:string, Company:string, UnitsSold:long)
[
    "New York", "Contoso", 300,
    "New York", "Litware", 1000,
    "New York", "Relecloud", 300,
    "New York", "Fabrikam", 200,
    "Seattle", "Contoso", 300,
    "Seattle", "Litware", 100,
    "Seattle", "Fabrikam", 100,
    "San Francisco", "Relecloud", 500,
    "San Francisco", "Litware", 500,
    "Washington DC", "Litware", 300,
    "Washington DC", "Contoso", 400
]
| sort by Location desc, UnitsSold desc
| extend Rank=row_rank_dense(UnitsSold, prev(Location) != Location)
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements

Yes

No

Litware from New York will be displayed at the top of the result set.

☐
☐

Fabrikam in Seattle will have value = 2 in the Rank column.

☐
☐

Litware in San Francisco will have the same value in the Rank column as Litware in New York.

☐
☐

A. See Explanation section for answer.

Answer(s): A

Explanation:

Answer Area

Statements

Litware from New York will be displayed at the top of the result set.

Yes



No



Fabrikam in Seattle will have value = 2 in the Rank column.



Litware in San Francisco will have the same value in the Rank column as Litware in New York.



Explanation:

Litware from New York will be displayed at the top of the result set - Yes The data is sorted first by Location in descending order and then by UnitsSold in descending order. Since "New York" is alphabetically the last Location, it will appear first in the result set. Within "New York", Litware has the highest UnitsSold (1000), so it will be displayed at the top.

Fabrikam in Seattle will have value = 2 in the Rank column - No The row_rank_dense function assigns dense ranks based on UnitsSold within each location. In "Seattle":

Contoso has UnitsSold = 300 Rank 1

Litware has UnitsSold = 100 Rank 2

Fabrikam also has UnitsSold = 100, so it shares the same rank (2) as Litware.

Litware in San Francisco will have the same value in the Rank column as Litware in New York - No The rank is calculated separately for each location. In "San Francisco":

Both Relecloud and Litware have UnitsSold = 500, so they share the same rank (1).

In "New York", Litware has the highest UnitsSold = 1000 Rank 1.

Since ranks are calculated independently for each location, Litware in San Francisco does not share the same rank as Litware in New York.

QUESTION: 110

You have a Fabric workspace that contains a lakehouse named Lakehouse1. Lakehouse1 contains a Delta table named Table1.

You analyze Table1 and discover that Table1 contains 2,000 Parquet files of 1 MB each.

You need to minimize how long it takes to query Table1.

What should you do?

A. Disable V-Order and run the OPTIMIZE command.

B. Disable V-Order and run the VACUUM command.

C. Run the OPTIMIZE and VACUUM commands.

Answer(s): C

Explanation:

Problem Overview:

Table1 has 2,000 small Parquet files (1 MB each).

Query performance suffers when the table contains numerous small files because the query engine must process each file individually, leading to significant overhead.

Solution:

To improve performance, file compaction is necessary to reduce the number of small files and create larger, optimized files.

Commands and Their Roles:

OPTIMIZE Command:

- Compacts small Parquet files into larger files to improve query performance.
- It supports optional features like V-Order, which organizes data for efficient scanning.

VACUUM Command:

- Removes old, unreferenced data files and metadata from the Delta table.
- Running VACUUM after OPTIMIZE ensures unnecessary files are cleaned up, reducing storage overhead and improving performance.

QUESTION: 111

You have a Fabric workspace that contains a warehouse named Warehouse1. Data is loaded daily into Warehouse1 by using data pipelines and stored procedures.

You discover that the daily data load takes longer than expected.

You need to monitor Warehouse1 to identify the names of users that are actively running queries.

Which view should you use?

- A. sys.dm_exec_connections
- B. sys.dm_exec_requests
- C. queryinsights.long_running_queries
- D. queryinsights.frequently_run_queries
- E. sys.dm_exec_sessions

Answer(s): E

Explanation:

sys.dm_exec_sessions provides real-time information about all active sessions, including the user, session ID, and status of the session. You can filter on session status to see users actively running queries.

QUESTION: 112

You have a Fabric workspace that contains an eventstream named EventStream1. EventStream1 outputs events to a table in a lakehouse.

You need to remove files that are older than seven days and are no longer in use.

Which command should you run?

- A. VACUUM
- B. COMPUTE
- C. OPTIMIZE
- D. CLONE

Answer(s): A

Explanation:

VACUUM is used to clean up storage by removing files no longer in use by a Delta table. It removes old and unreferenced files from Delta tables. For example, to remove files older than 7 days:

VACUUM delta.`/path\_to\_table` RETAIN 7 HOURS;

QUESTION: 113

You have a Fabric warehouse named DW1 that loads data by using a data pipeline named Pipeline1.

Pipeline1 uses a Copy data activity with a dynamic SQL source. Pipeline1 is scheduled to run every 15 minutes.

You discover that Pipeline1 keeps failing.

You need to identify which SQL query was executed when the pipeline failed.

What should you do?

- A. From Monitoring hub, select the latest failed run of Pipeline1, and then view the output JSON.
- B. From Monitoring hub, select the latest failed run of Pipeline1, and then view the input JSON.

- C. From Real-time hub, select Fabric events, and then review the details of Microsoft.Fabric.ItemReadFailed.
- D. From Real-time hub, select Fabric events, and then review the details of Microsoft.Fabric.ItemUpdateFailed.

Answer(s): B

Explanation:

The input JSON contains the configuration details and parameters passed to the Copy data activity during execution, including the dynamically generated SQL query.

Viewing the input JSON for the failed pipeline run provides direct insight into what query was executed at the time of failure.

QUESTION: 114

You have a Fabric notebook named Notebook1 that has been executing successfully for the last week.

During the last run, Notebook1 executed nine jobs.

You need to view the jobs in a timeline chart.

What should you use?

- A. Real-Time hub
- B. Monitoring hub
- C. the job history from the application run
- D. Spark History Server
- E. the run series from the details of the application run

Answer(s): E

Explanation:

The run series from the details of the application run is the most detailed and relevant feature for visualizing job execution in a timeline format, making it the correct choice for this scenario. It provides an intuitive way to analyze job execution patterns and improve the efficiency of the notebook.

QUESTION: 115

HOTSPOT (Drag and Drop is not supported)

You have a Fabric workspace that contains an eventstream named EventStream1.

You discover that an EventStream1 transformation fails.

You need to find the following error information:

The error details, including the occurrence time

The total number of errors

What should you use? To answer, select the appropriate options in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

To find the error details:

	▼
Data insights	
Data preview	
Details	
Runtime logs	

To find the total number of errors:

	▼
Data insights	
Data preview	
Details	
Runtime logs	

A. See Explanation section for answer.

Answer(s): A**Explanation:****Answer Area**

To find the error details:

	▼
Data insights	
Data preview	
Details	
Runtime logs	

To find the total number of errors:

	▼
Data insights	
Data preview	
Details	
Runtime logs	

Explanation:

To Find the Error Details - Runtime logs

Runtime logs provide detailed information about the execution of transformations, including specific errors that occurred, their descriptions, and the exact timestamp when the error happened. This is where you can get the most comprehensive details regarding failures, including the occurrence time.

Runtime logs are used to gather detailed information about individual errors, including descriptions and timestamps.

To Find the Total Number of Errors - Details

The Details section typically provides a summary view of the operation, including statistics such as the

total number of errors encountered during the execution of an eventstream or data pipeline. This allows you to easily see the error count without digging into individual logs. The Details section provides a summary or overview, including the number of errors that occurred during a transformation or operation.

QUESTION: 116

You have a Fabric workspace that contains a warehouse named Warehouse1. While monitoring Warehouse1, you discover that query performance has degraded during the last 60 minutes. You need to isolate all the queries that were run during the last 60 minutes. The results must include the username of the users that submitted the queries and the query statements. What should you use?

- A. the Microsoft Fabric Capacity Metrics app
- B. views from the queryinsights schema
- C. Query activity
- D. the sys.dm_exec_requests dynamic management view

Answer(s): C

Explanation:

Monitor your running and completed T-SQL queries using Query activity Monitoring SQL queries is essential for monitoring and troubleshooting performance of your Fabric warehouse, and also for maintaining the efficiency of the warehouse. With Query activity, you have a one-stop view of all running and historical T-SQL queries along with a list of long-running and frequently run queries without having to run any T-SQL code. You can use Query activity ensure that queries aren't taking longer than expected to execute and are completing successfully.

Query runs

On the Query runs page, you can see a list of running, succeeded, canceled, and failed queries up to the past 30 days.

Use the dropdown list to filter for status, submitter, or submit time.

Use the search bar to filter for specific keywords in the query text or other columns.

QUESTION: 117

You have a Fabric workspace that contains a semantic model named Model1. You need to monitor the refresh history of Model1 and visualize the refresh history in a chart. What should you use?

- A. the refresh history from the settings of Model1
- B. a notebook
- C. a Dataflow Gen2 dataflow
- D. a data pipeline

Answer(s): B

Explanation:

Semantic Link for automation

Microsoft Fabric Notebooks can work with Fabric Semantic Link. In short, Semantic Link allows you to do all sorts of operations with Power BI Semantic Models ranging from reading the meta data, data itself to automating various activities such as refreshes.

QUESTION: 118

You have a Fabric workspace that contains a write-intensive warehouse named DW1. DW1 stores staging tables that are used to load a dimensional model. The tables are often read once, dropped, and then recreated to process new data.

You need to minimize the load time of DW1.

What should you do?

- A. Enable V-Order.
- B. Create statistics.
- C. Drop statistics.
- D. Disable V-Order.

Answer(s): D

Explanation:

V-Order is a write time optimization to the parquet file format that enables lightning-fast reads under the Microsoft Fabric compute engines, such as Power BI, SQL, Spark, and others.